



MSc Thesis for Health Data Science

A Bayesian time series Framework for Comparing On Trolley Counts Across Irish HSE Health Regions

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Declaration

We hereby certify that this material, which we now submit for assessment on the programme of study leading to the award of Master of Science is entirely our own work and had not been taken from the work of others or Artificial Intelligence, save and to the extent that such work has been cited and acknowledged within the text of our work.

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Contents

1	Introduction	4
2	Literature Review	5
3	Data	7
3.1	Data Sources	7
3.2	Scraping	7
3.3	Preprocessing	7
3.4	Exploratory Data Analysis	8
3.4.1	New Year Dip	8
3.4.2	Mid West Reset	8
3.4.3	Spectral Analysis	9
3.4.4	Autocorrelation	10
4	Methods	11
4.1	Response Scaling	11
4.1.1	Per 10,000 population	11
4.1.2	Per €1 billion of regional budget	12
4.1.3	Per 100 inpatient beds	12
4.1.4	Per 1,000 population aged 65+	12
4.2	Model Comparison	12
4.3	Model Development	12
4.4	Bayesian Inference	14
4.4.1	Likelihood	14
4.4.2	Priors	14
4.4.3	Posterior	15
4.5	Bayesian Ranking	15
4.6	Pairwise tests	15
5	Results	16
5.1	Mean Component	16
5.1.1	Regional Levels	16
5.2	Regional Rankings	17
5.2.1	Per 10,000 Population	17
5.2.2	Sensitivity to Scaling	17
5.3	Annual Cycle	18
5.3.1	Annual Cycle Phase	18
5.3.2	Annual Cycle Amplitude	19
5.4	New Year Effect	20
5.5	Mid West Reset	20
5.6	Model Extensions	20
5.7	Residuals	21
5.7.1	Cross-Correlation	21
6	Discussion	22
7	Limitations and Future Work	24
8	Conclusion	25
9	References	26
10	Appendix	28
10.1	Annual Cycle Peak Phase	28

10.2	Model Development DIC	28
10.3	Results of Regional Structure Changes	28
10.4	Pairwise Regional Level Comparisons	28
10.5	Pairwise Cycle Amplitude Comparisons	29
10.6	Raw Weekly Counts by Region	30
10.7	Residual Normality Checks	30
10.8	Residual Cycles	31
10.9	Trace Plots and Convergence	32

1 Introduction

Hospital overcrowding has become a persistent challenge for healthcare systems worldwide (Morley et al., 2018). When hospitals operate at or beyond capacity, the consequences extend beyond operational strain. Overcrowding in emergency departments is associated with increased patient mortality, delayed time-critical treatments, longer waiting times, and reduced quality of care (Asplin et al., 2003). Understanding and monitoring hospital load is therefore not only a resource management concern but a patient safety issue.

In Ireland, hospital congestion is commonly monitored using the number of patients recorded on hospital trolleys. These patients have been admitted through emergency departments but remain without an available inpatient bed, often receiving care in temporary spaces such as corridors or emergency wards. Trolley counts are widely used as a daily indicator of hospital system load and are frequently referenced in public reporting and policy discussions concerning hospital capacity and healthcare performance (Health Service Executive, 2026a).

The Health Service Executive publishes daily trolley counts across hospitals nationwide. Despite their widespread use, these counts are typically presented as raw daily totals, offering limited insight into the underlying dynamics of hospital demand. Raw counts do not adjust for population size, meaning regions with larger populations will naturally record higher totals regardless of the relative load on their systems. They also do not account for temporal patterns such as seasonal fluctuations in illness demand, nor do they allow systematic or probabilistic comparisons between healthcare regions.

Ireland’s six HSE health regions differ substantially in population size, age structure, and acute hospital capacity. The three eastern Dublin regions, Dublin and North East (1.19 million), Dublin and Midlands (1.08 million), and Dublin and South East (0.97 million), are the most populous and carry the largest concentrations of inpatient beds, with Dublin and North East alone accounting for 2,814 available beds (Central Statistics Office, 2024a; Health Service Executive, 2025). The three western regions, West and North West (760,000), South West (741,000), and Mid West (413,000), serve smaller and more dispersed populations across predominantly rural catchments, with acute services concentrated in fewer centres (Central Statistics Office, 2024a). These structural differences are compounded by age: people aged 65 and over account for approximately 15% of the national population but 40% of hospital discharges, and the share of older residents varies across regions, with the western regions carrying a proportionally higher burden (Brick & Kakoulidou, 2025; Central Statistics Office, 2023, 2024b). Taken together, these demographic and capacity differences mean that raw hospital activity counts are poorly suited to cross-regional comparison, as a region with a larger population will naturally record higher absolute counts regardless of the relative pressure on its system, making per capita standardisation essential for any meaningful assessment of regional hospital load. Figure 1 summarises these contrasts across the six regions: population, available inpatient beds, budget, and population aged 65 and over.

Hospital demand is influenced by multiple factors including population demographics, seasonal illness patterns, healthcare infrastructure, and patterns of emergency department admissions. These factors generate complex temporal dynamics in hospital congestion that cannot be fully understood through descriptive statistics alone. Policy responses such as capacity management protocols, discharge planning initiatives, and operational resets further shape observed demand patterns, making it important to account for known interventions when modelling trolley counts over time. Hospital demand and the congestion it generates are inherently temporal: counts recorded in one week are likely to be influenced by counts in preceding weeks, congestion tends to persist until patients are discharged or capacity is increased, and seasonal patterns in respiratory illness mean that load is not uniform across the year. time series modelling provides a useful framework for analysing such data, as it allows researchers to account for these sources of temporal dependence and recurring structure, enabling the underlying level of each region to be estimated more reliably and systematic patterns to be separated from week-to-week noise (Shumway & Stoffer, 2017).

Bayesian statistical methods provide a natural framework for incorporating uncertainty into such analyses. Rather than producing single point estimates, Bayesian models treat parameters as probability distributions and update these using observed data, producing probabilistic estimates that are often more informative than traditional approaches (Gelman et al., 2013). The Bayesian ranking approach of Laird & Louis (1989) extends this to produce probabilistic orderings of units that account for uncertainty in the underlying parameter

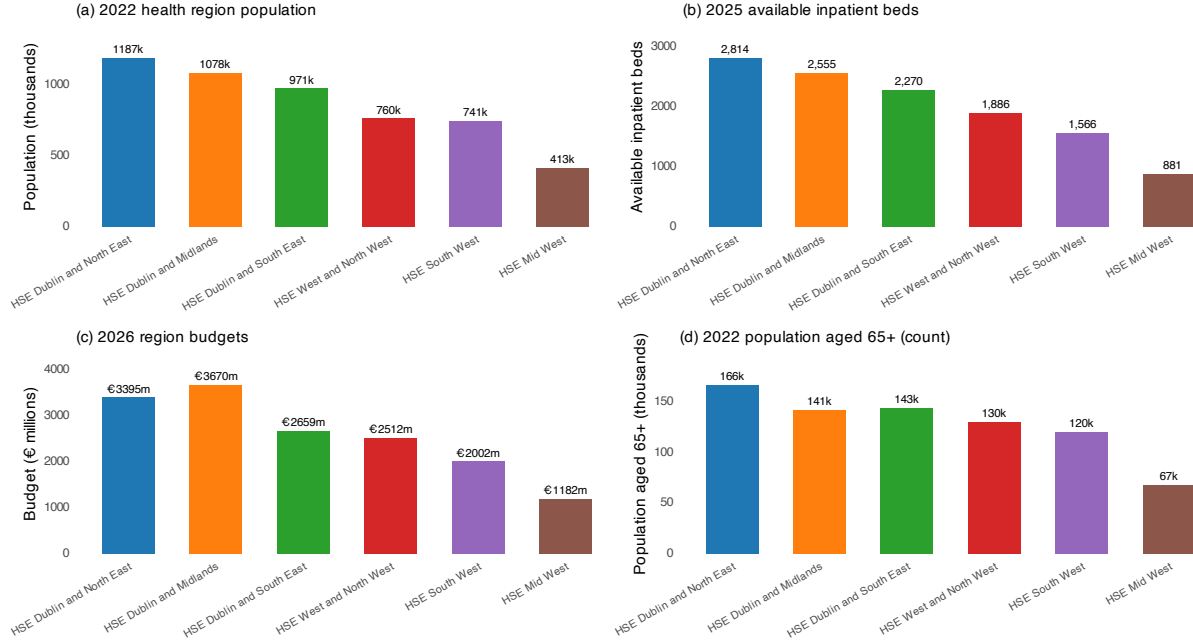


Figure 1: HSE health region characteristics: (a) 2022 health region population, (b) 2025 available inpatient beds, (c) 2026 region budgets, (d) 2022 count of population aged 65+ (thousands). Regions are ordered by population (largest first) and shared across panels so each region occupies the same column.

estimates, making it particularly well suited to regional comparison in healthcare settings.

The HSE organises its hospital network into six health regions: Dublin and Midlands, Dublin and North East, Dublin and South East, Mid West, South West, and West and North West. Daily trolley counts at hospital level have been publicly available since January 2023 (Health Service Executive, 2026a), providing a consistent basis for regional comparison over time. This study develops a Bayesian time series framework for comparing hospital trolley load across these six regions, using weekly trolley counts from January 2023 to April 2026 sourced from the HSE Urgent and Emergency Care daily report. Trolley counts are standardised by regional population and expressed as rates per 10,000 residents to allow meaningful cross-regional comparison. The modelling framework decomposes regional trolley rates into a regional level, an annual cycle, New Year effects, and where applicable, the impact of known operational interventions, with an autoregressive component capturing week-to-week persistence. A notable feature of the data is the HSE Mid West operational reset in August 2024, during which scheduled care was deferred to address acute overcrowding, and the modelling framework is designed to isolate the effect of such events from the underlying annual cycle and level.

The primary objective of this research is to provide a statistically rigorous basis for comparing hospital load across Irish HSE regions, identify the temporal and operational features driving trolley counts, and demonstrate that Bayesian time series modelling of routinely collected administrative data can yield policy relevant insights to support capacity planning and resource allocation decisions across the Irish health system.

2 Literature Review

Hospital overcrowding and its consequences have been extensively documented in the international literature. Overcrowding in emergency departments is associated with increased patient waiting times, reduced quality of care, and greater strain on healthcare staff (Asplin et al., 2003; Morley et al., 2018). In Ireland, this issue has been persistent and politically prominent, with trolley counts serving as the primary public metric of hospital system pressure for over two decades. At a descriptive level, persistent high trolley counts in western hospitals, particularly University Hospital Limerick and University Hospital Galway, have been documented repeatedly in INMO Trolley Watch reports, with both hospitals consistently ranking

among the most overcrowded in the country (Bowers, 2026; Irish Nurses and Midwives Organisation, 2024). However, these raw counts have not previously been subjected to a formal regional comparison that controls for population size, seasonal dynamics, and known operational events.

A well-established driver of emergency demand in Ireland is seasonal respiratory illness. Mulroe et al. (2019) analysed adult emergency hospitalisations nationally from 2015 to 2018 and found that while overall emergency admissions were only marginally higher in winter, respiratory conditions increased by 40.5% relative to summer, driven primarily by chronic obstructive pulmonary disease, pneumonia, and lower respiratory tract infection. This winter surge is consistent with patterns documented internationally and provides direct epidemiological motivation for modelling seasonal structure explicitly rather than treating it as noise.

Internationally, Bayesian and time series methods have been applied to hospital demand data across a range of settings, though the predominant focus has been short-term operational forecasting at individual hospital sites rather than sustained regional comparison. Sahu et al. (2014) develop a hierarchical Bayesian model for short-term prediction of hospital admissions in Cardiff and Southampton, incorporating seasonal and meteorological variables and propagating forecast uncertainty through the posterior distribution; this work is methodologically relevant to the present study but is oriented towards prediction rather than characterisation of long-run regional differences. Upshur et al. (2005) apply time series methods to population-based hospital admission data across the province of Ontario from 1988 to 2001, finding systematic seasonal patterns across 33 of the 52 most common admission diagnoses; their work establishes that hospital admissions have predictable seasonal structure that can be modelled formally, though it does not extend to probabilistic regional comparison or Bayesian inference. More recently, Mellor et al. (2023) use hierarchical generalised additive models to forecast influenza hospital admissions across English sub-regions, demonstrating that hierarchical structures can exploit shared information across geographic units to improve forecasts. While these studies collectively establish the methodological foundations on which the present study draws, none combines Bayesian time series decomposition with population-standardised rates and probabilistic regional ranking applied to routinely collected administrative data over a sustained multi-year period. The closest parallel in spirit is the work of Jacob et al. (2024), who find substantial residual variation of the order of three to one in per-person emergency department attendance rates across local areas in England after controlling for population risk factors, demonstrating that raw counts systematically obscure underlying regional disparities, a finding that directly motivates the population-standardised approach adopted here.

A common approach for capturing annual cycles in hospital data is trigonometric regression, in which cosine and sine terms at the relevant frequency are included as fixed covariates within a regression or state-space framework, allowing both the amplitude and phase of the cycle to be estimated from the data (Shumway & Stoffer, 2017). This specification has practical advantages over dummy-variable approaches: it uses two parameters to represent a full annual cycle regardless of the length of the series, and the amplitude and peak timing are directly interpretable from the estimated coefficients. The present study adopts this approach within a Bayesian framework.

Bayesian methods have an established tradition in hospital performance profiling and regional comparison. Laird & Louis (1989) introduced an empirical Bayes ranking approach that treats unit-level parameters as draws from a common prior, producing probabilistic rank distributions that account for estimation uncertainty. This approach has subsequently been applied to hospital profiling, where ranking institutions on a single metric risks overstating certainty in orderings that may be largely driven by chance variation (Austin, 2002). MacDermott et al. (2025) demonstrate the practical value of the Laird and Louis framework in a comparative performance setting, applying Bayesian ranking to Olympic medal table data and showing that credible intervals on rank positions frequently span several places, illustrating that apparent differences in performance may not be statistically credible. The present study applies this framework to regional hospital trolley rates, where the question of whether observed differences in burden reflect genuine structural disparities or sampling noise is of direct policy relevance.

Within the Irish context, the most directly relevant study is Lynch et al. (2019), who apply a structural breaks time series analysis to weekly trolley counts across four HSE regions from 2005 to 2015, finding significant post-reconfiguration changes in trolley numbers in three of the four regions studied. This is the only published time series analysis of Irish trolley data at a regional level that we are aware of. However, the

data used are sourced from INMO Trolley Watch reports, which count differently from the HSE report used here by including surge-bed and ward overflow patients the HSE excludes (RTÉ, 2023). They further exclude weekends and bank holidays, cover only four regions under the pre-2023 HSE structure, and are not population standardised, limiting the comparability of pressure levels across regions. The frequentist interrupted time series framework is furthermore designed to evaluate the effect of a specific policy intervention rather than to characterise underlying temporal dynamics or support systematic regional comparison with quantified uncertainty.

These gaps motivate the present study. No published analysis has applied a Bayesian time series framework to characterise and compare long-run hospital pressure across Irish health regions with explicit uncertainty quantification, nor has any study examined trolley data under the current six-region HSE structure established in 2023. The present work addresses both gaps by developing a Bayesian time series model that estimates region-specific pressure levels, accounts for seasonal and calendar dynamics, and enables principled probabilistic comparison of regions on a population-standardised basis.

3 Data

This section outlines the data used in the analysis: collecting the raw data, preparing it for modelling, and exploring the series to identify important features.

3.1 Data Sources

Daily on trolley counts are scraped from the publicly available HSE Urgent and Emergency Care daily report (Health Service Executive, 2026a). The report is published as an HTML page with no known API. It has been published daily since 1st of January 2023 and remains actively updated. For our paper, we collected data from the 1st of January 2023 to the 5th of April 2026. The HSE Urgent and Emergency Care daily report contains counts of patients on ED trolleys, ward trolleys, surge capacity in use, delayed transfers of care, the total number waiting over 24 hours, and the number of those over 75 waiting over 24 hours. Our analysis uses the total on trolley counts, a sum of ED and ward trolleys. Total counts are chosen because they are the most direct measure of the burden of patients on trolleys in each region, and they are a widely reported measure of hospital crowding (Health Service Executive, 2026a). All code for scraping, preprocessing, model fitting, and figure generation is available at <https://github.com/hotsoupisgood/hse-urgent-care-report-trolley-analysis>, supporting full reproduction of the results.

3.2 Scraping

Scraping is performed using Python, iterating through each day’s report from the 1st of January 2023 to the 5th of April 2026. The report is scraped from the URL: <https://uec.hse.ie/uec/TGAR.php?EDDATE=DD%2FMM%2FYYYY>, where EDDATE is the date in DD/MM/YYYY (day/month/year) format, with %2F being encoded slashes. On this report page, data is displayed in the `<table>` tag, which produces a structured HTML table, where rows are coded inside `<tr>` tags, and cells are coded inside `<td>` tags. The scraper extracts the date, region, and total daily on trolley count from the table, and saves it to a CSV file for further processing covered in the following section.

3.3 Preprocessing

Preprocessing involves validating, cleaning, and transforming the raw daily hospital counts into the weekly regional counts. The daily counts are summed into weekly counts, with Monday designated as the first day of the week. Hospital counts are summed into regional weekly totals and a final CSV is produced which is explored throughout the rest of the analysis.

3.4 Exploratory Data Analysis

The first step in exploring the preprocessed weekly trolley counts is to identify key patterns in the time series. The second step is to gather further evidence for the patterns with more rigorous plots (Figures 3, 4, 5, 7). In Figure 2 each region is a different coloured line. The horizontal axis is time from initial release of the report on the 1st of January 2023 to the 5th of April 2026. The Figure suggests three key features: an annual cycle peaking near winter; a sharp dip each New Year, marked by light blue bands; and an isolated dip in HSE Mid West during August 2024, marked by the light red band (the 8th to 18th of August window). Each feature is examined with supporting evidence and sources in the following paragraphs.

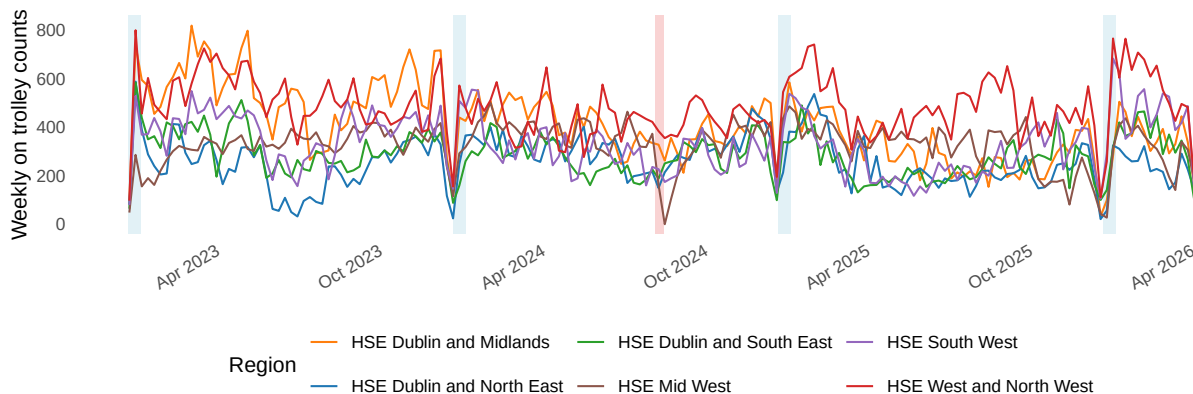


Figure 2: Weekly total on trolley counts by HSE region from 2023 to 2026. Light blue bands mark the New Year weeks across all regions; the light red band marks the Mid West reset window (8th to 18th of August 2024).

3.4.1 New Year Dip

An annual dip around the New Year aligns with the HSE practice of discharging patients over Christmas to prepare for a January surge driven by increased respiratory illness (Department of Health, 2023) and delayed hospital visits (Health Service Executive, 2026b). Figure 3 shows the New Year dip in HSE Dublin and Midlands (an example region) for 2024, 2025, and 2026 as traces. The dashed line at week 0 marks the 1st of January, and each week is positioned by its starting date as a point on the graph. A consequence of weekly aggregation is the 1st of January falls on a different weekday each year. So the New Year week rate is split unevenly across adjacent weekly bins, which can lead to misinterpretation of the data. The plot shows that the New Year reaches a minimal level approximately 1 to 2 weeks before the New Year week. Notably, 2026 has the lowest dip in the series; the remaining regions follow the same pattern, except HSE Dublin and South East where 2026 has the highest dip across available years.

3.4.2 Mid West Reset

In contrast to the New Year dip, the Mid West dip is not observed in other regions, and does not align with any known cycle. The red shaded period in Figure 4 shows the HSE Mid West region during what has been reported as the reset period. In response to high demand, HSE Mid West delayed scheduled care to focus on unscheduled care from the 8th to 18th of August 2024 (Health Information and Quality Authority, 2025). The reset period overlaps with the dip in on trolley counts in the Mid West region as seen in Figure 4. This dip is not observed in the other regions, seen by comparing the non-Mid West grey lines in Figure 4 with the Mid West red line, suggesting the dip is from the reset rather than an effect common to all regions. The New Year dip and Mid West reset both lack obvious parametric forms, and will require a block effect model which will be discussed in Section 4.3. In the following sections we will examine the spectral and autocorrelation properties of the series to determine how to capture these parametric effects.

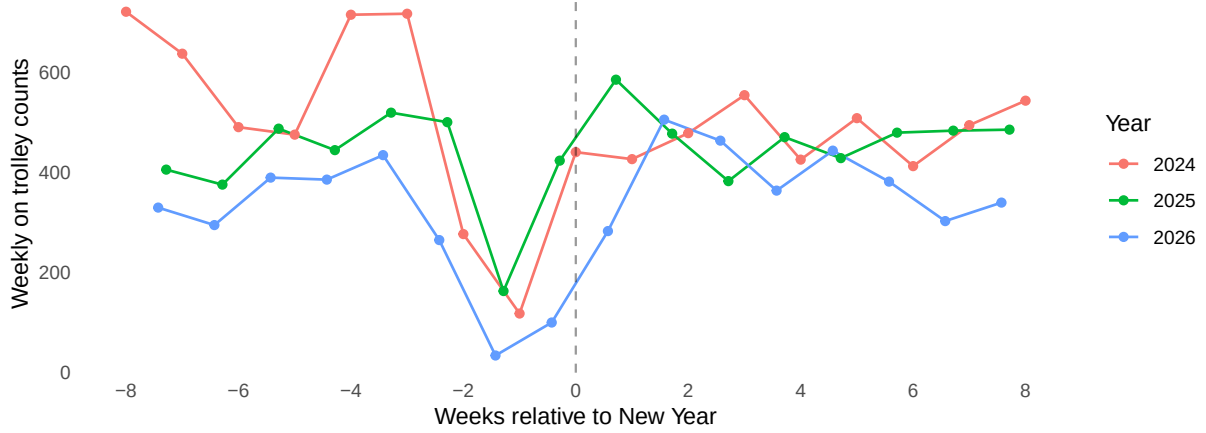


Figure 3: New Year for HSE Dublin and Midlands, weekly total trolleys in the 8 weeks either side of the 1st of January for each year in the series.

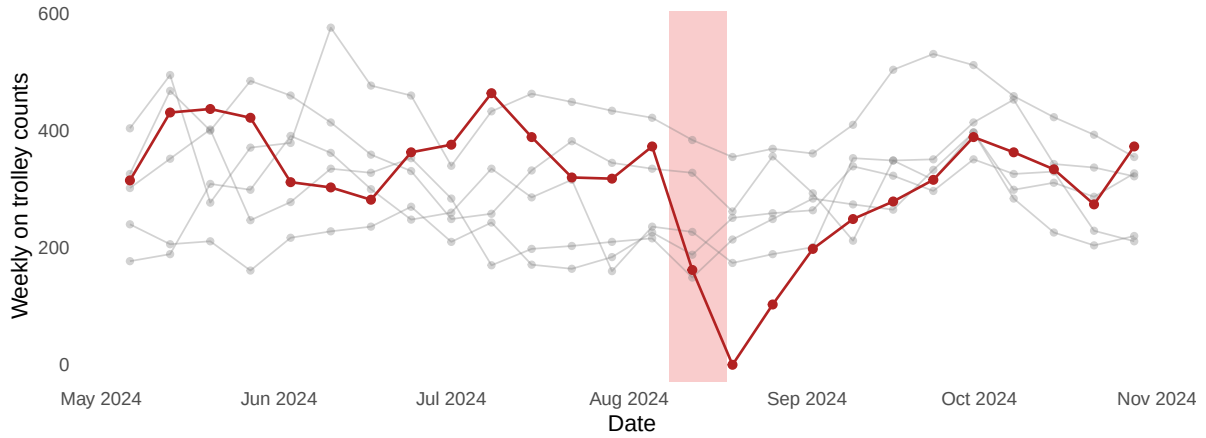


Figure 4: HSE Mid West weekly trolley totals from May to November 2024. The shaded period marks the 8th to 18th of August Mid West reset period. The grey lines are non-Mid West regions over the same time span.

3.4.3 Spectral Analysis

The spectral analysis is performed using a power spectral density (PSD), estimated by the periodogram below. The Fourier transform represents a signal as a sum of sinusoids at particular frequencies, each with an amplitude and phase (Smith, 1999). The PSD reports the power, the squared amplitude, at each frequency and discards phase. The transform is computed via the fast Fourier transform (FFT). On the vertical axis, high power indicates a strong cycle at that frequency. The periodogram in Figure 5 shows high power particularly in the 52 week period, and also heightened power near the 1-month, 2-month, and 6-month marks (4, 8, and 26 weeks).

The regions form two groups by their power at the annual cycle. HSE South West, Dublin and North East, and Dublin and Midlands group with the highest power, and at the lowest power HSE Dublin and South East, Mid West, and West and North West group together. The annual cycle is the most notable cycle in the series. But given we only have three full years of New Year data (2024, 2025, and 2026) any estimates of the annual cycle will be imprecise.

Beyond the annual cycle, the periodogram shows weaker peaks just off the 1-month, 2-month, and 6-month marks (4, 8, and 26 weeks). These peaks may not centre on their true period due to the limited number of data points. They could represent monthly, 2-month, or 6-month cycles. A 6-month cycle could be influenced by hospital doctor (NCHD) changeovers in January and July (PE Global, 2026). They could also

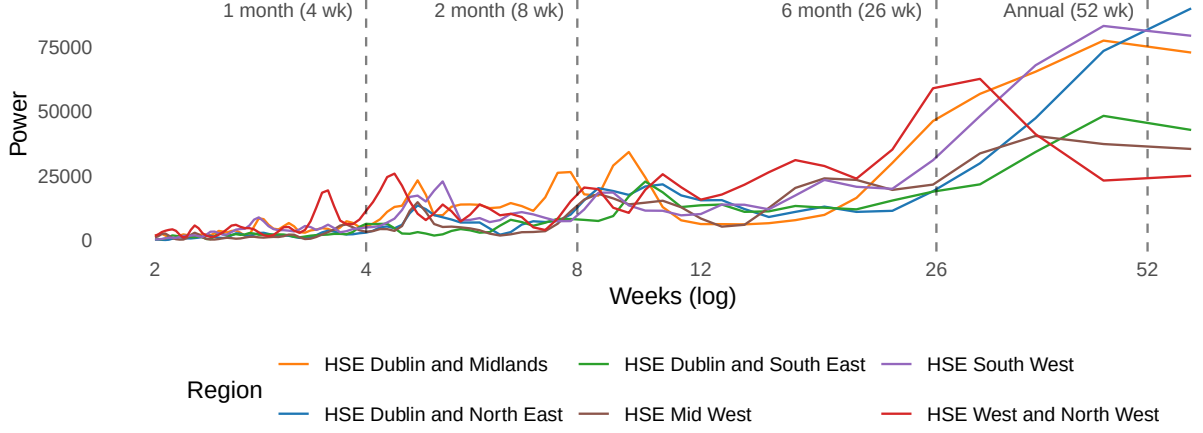


Figure 5: Periodogram of weekly trolley counts for each HSE region. Dashed lines mark reference periods at 4, 8, 26, and 52 weeks. The horizontal axis is the period in weeks on a log scale.

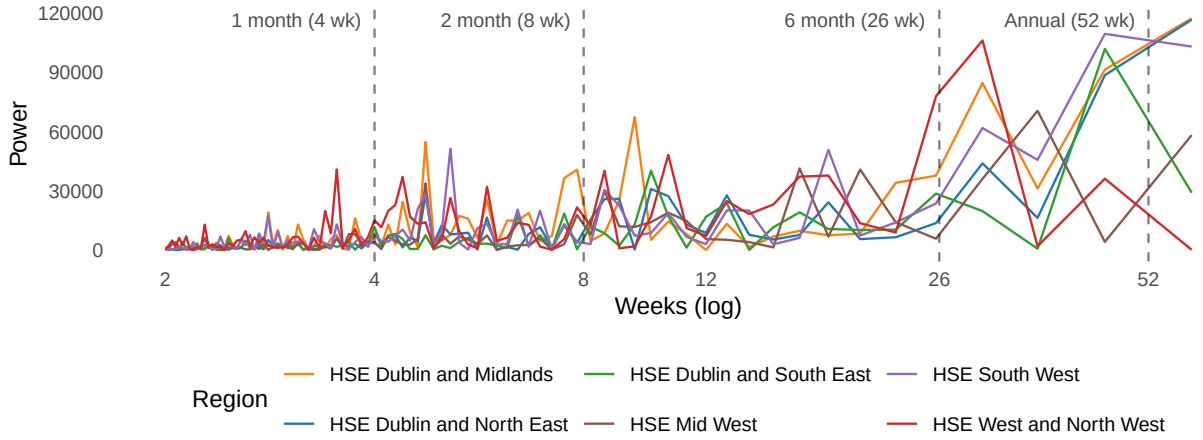


Figure 6: Raw (unsmoothed) periodogram of weekly trolley counts for each HSE region. Dashed lines mark reference periods at 4, 8, 26, and 52 weeks. The horizontal axis is the period in weeks on a log scale.

be harmonics of the annual cycle.

The 6-month peak is clearest in HSE West and North West and Dublin and Midlands. At the 2-month mark HSE Dublin and Midlands has the highest peak. HSE West and North West, Dublin and North East, Mid West, and South West also peak in this area. The 1-month peak is the weakest. HSE West and North West and Dublin and Midlands are strongest and HSE Dublin and South East is weakest.

These peaks do not sit on the 1-month, 2-month, or 6-month marks. They fall closer to the harmonics of the annual cycle, which lie at $52/n$ weeks (26, 8.7, and 4.3 weeks), than to the calendar months. This favours harmonics of a non-sinusoidal annual cycle over genuine monthly cycles. The evidence is still not clear. Three years of weekly data may place the peaks imprecisely, so the periodogram ultimately cannot confirm a sub-annual cycle. No conclusion is drawn from the spectrum alone. Each candidate cycle is tested as a model component and compared by DIC (Table 3).

3.4.4 Autocorrelation

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are used to analyse the temporal dependencies in the weekly trolley counts. Where the ACF is $\rho_k = \text{corr}(x_t, x_{t-k})$ and the PACF is $\phi_{kk} = \text{corr}(x_t, x_{t-k} \mid x_{t-1}, \dots, x_{t-k+1})$, the correlation at lag k with the intermediate lags partialled out. In Figure 7 in the vertical axis is the autocorrelation, the horizontal axis is the lag in weeks, and the dashed lines

mark the 95% confidence band; bars beyond it are significant. Figure 7 shows the ACF (a) and PACF (b) for HSE Dublin and Midlands. The ACF decays slowly and remains significant across many lags, consistent with an annual cycle. The PACF has a large lag 1 value that drops off sharply, indicating an AR(1) effect. The PACF also has significant peaks at 3, 4, and 5 weeks indicating an additional monthly cycle.

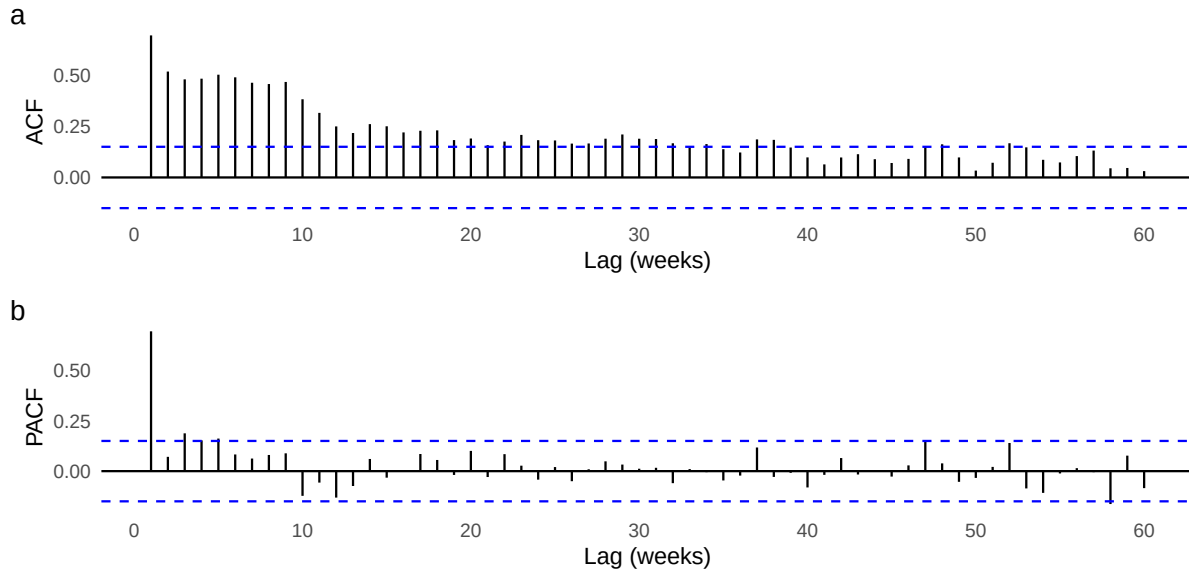


Figure 7: (a) Autocorrelation and (b) partial autocorrelation functions of weekly trolley totals for HSE Dublin and Midlands. Dashed lines mark the 95% confidence band.

4 Methods

This section covers the four steps that take the explored series to a regional ranking: scaling the response (Section 4.1), developing the model (Section 4.3), fitting it with Bayesian inference (Section 4.4), and ranking the regions. The primary scale throughout is the rate per 10,000 population.

4.1 Response Scaling

Ranking regions by the raw on trolley counts would be misleading, as ranks would primarily reflect the population size of each region. For example, HSE Dublin and North East has the largest population and the highest raw on trolley counts, but this does not necessarily mean it is performing worse than other regions. To investigate other values contributing to our ranking, several denominators are used to scale the response. Scaling allows comparisons of the regions on an equal level, as the raw counts are influenced by factors such as population size, funding and bed capacity. The per 10,000 population scaling is the primary measure used throughout, as it most directly controls for differences in region size. Sensitivity to the alternative denominators (budget, beds, and population aged 65+) is examined in Section 5.2.2.

4.1.1 Per 10,000 population

The primary denominator used is the population of each health region in 2022. The weekly on trolley count is then divided by this population and scaled to a rate per 10,000 population:

$$y_{j,t} = \frac{\text{weekly trolley count}_{j,t}}{\text{health region population}_j} \times 10,000.$$

Per 10,000 population is the primary scaling as it is the most direct way to control for differences in demand between regions. So our results default to this scaling. The scales below largely follow similar

distributions to population size.

4.1.2 Per €1 billion of regional budget

To control for funding differentials, on trolley counts are scaled by the 2026 HSE regional budget allocation, expressing the outcome as trolleys per €1 billion of regional budget.

$$y_{j,t} = \frac{\text{weekly trolley count}_{j,t}}{\text{health region budget in billion (€)}_j}.$$

Regions with lower on trolley counts per €1 billion of regional budget may be considered more efficient.

4.1.3 Per 100 inpatient beds

A similar efficiency measure is applied to inpatient beds in each region in 2025. The number of inpatient beds is used as a proxy for the space capacity of each region. Regions that have a lower on trolley count per 100 inpatient beds may be considered more efficient at handling patients within their allocated space. The outcome is expressed as trolleys per 100 inpatient beds:

$$y_{j,t} = \frac{\text{weekly trolley count}_{j,t}}{\text{number of health region inpatient beds}_j} \times 100.$$

4.1.4 Per 1,000 population aged 65+

People aged 65+ disproportionately utilise hospitals (Brick & Kakoulidou, 2025). 65+ make up 15% of the population but 40% of discharges (Central Statistics Office, 2023, 2026). Scaling by region population aged 65+ controls for differences in age structure between regions (Central Statistics Office, 2024a, 2024b).

$$\text{population aged 65+}_j = \text{health region population}_j \times \frac{\% \text{ aged 65+}_j}{100}.$$

The scaling is then expressed as trolleys per 1,000 population aged 65+:

$$y_{j,t} = \frac{\text{weekly trolley count}_{j,t}}{\text{population aged 65+}_j} \times 1,000.$$

4.2 Model Comparison

Components are compared with the deviance information criterion (DIC) (Spiegelhalter et al., 2002). The DIC is used in the initial model development to support the observations from the exploratory analysis. The DIC ranks candidate models by their ability to predict new data of the same kind. It is the posterior mean deviance plus p_D , the effective number of parameters. The p_D term penalises complexity and protects against overfitting. A smaller DIC suggests better predictive ability. A single DIC value has no meaning on its own. It is read only relative to other models on the same data. In general, a component is retained when it lowers DIC.

4.3 Model Development

Using the scaled responses discussed in the previous section, the model is now specified. The model is developed iteratively with increasing complexity, as in Gelman (2004), adding components from EDA and residual patterns. Each observation is modelled as a regional mean $\mu_{j,t}$ plus two autoregressive terms, $\mathcal{E}_{j,t}^1$ and $\mathcal{E}_{j,t}^2$. The mean captures the level and cycle structure, and the AR terms capture residual autocorrelation at lag 1 and lag 2. Initialisation of the first two lags is achieved via:

$$y_{j,1} \sim \mathcal{N}(\mu_{j,1}, \tau_j^{-1}); \quad y_{j,2} \sim \mathcal{N}(\mu_{j,2} + \mathcal{E}_{j,2}^1, \tau_j^{-1}); \quad y_{j,t} \sim \mathcal{N}(\mu_{j,t} + \mathcal{E}_{j,t}^1 + \mathcal{E}_{j,t}^2, \tau_j^{-1}), \quad t > 2.$$

Here $y_{j,t}$ is the scaled trolley count for region j at week t , τ_j is the region-specific precision, and the AR terms use coefficients ϕ_1 and ϕ_2 shared across all regions.

The mean ($\mu_{j,t}$) component is composed of: a regional level (α_j), an annual cycle ($\mathcal{C}_{j,t}^{52}$), a regional New Year effect ($\mathcal{R}_{j,t}$), and the Mid West reset effect ($\mathcal{MW}_t$). The mean component is fitted for each region, j , and week, t : $\mu_{j,t} = \alpha_j + \mathcal{C}_{j,t}^{52} + \mathcal{R}_{j,t} + \mathbf{1}(j = \text{MW}) \mathcal{MW}_t$.

The regional level and annual cycle are motivated by the exploratory analysis. Regions sit at different levels across the series (Figure 2), so each region is given its own level α_j . Removing it from the model and using one global level raises DIC from 5336.9 to 5538.2 (Appendix Table 6), so the regional level is kept on DIC grounds. The series peaks in winter and the periodogram has a dominant 52 week peak (Figures 2, 5), so an annual cycle is added. Its amplitude and phase are allowed to vary by region. This is not chosen on DIC. A single shared cycle in fact has a slightly lower DIC of 5312.3 against 5336.9, with the same deviance. The region-specific cycle is kept because the regional amplitude and phase are the quantities of interest (Section 5.3.2). The small DIC cost buys the regional comparison at no loss of fit.

For this model, the annual cycle is a sine wave reformulated as a linear regression:

$$\mathcal{C}_{j,t}^{52} = A_j \cos\left(\frac{2\pi t}{52} + \phi_j\right) = \beta_j \cos\left(\frac{2\pi t}{52}\right) + \gamma_j \sin\left(\frac{2\pi t}{52}\right)$$

. Where $\beta_j = A_j \cos(\phi_j)$ and $\gamma_j = -A_j \sin(\phi_j)$. From the posterior of β_j and γ_j , the amplitude $A_j = \sqrt{\beta_j^2 + \gamma_j^2}$ and peak week $m_j = \tan^{-1}(\gamma_j/\beta_j)$.

Because there is no obvious parametric form, the regional New Year effect is modelled as a block effect spanning the week containing the 1st of January, the two weeks before, and one week after:

$$\begin{aligned} \mathcal{R}_{j,t} = & \delta_{j,t} \mathbf{1}(t = \text{New Year week} - 2) \\ & + \delta_{j,t} \mathbf{1}(t = \text{New Year week} - 1) \\ & + \delta_{j,t} \mathbf{1}(t = \text{New Year week}) \\ & + \delta_{j,t} \mathbf{1}(t = \text{New Year week} + 1). \end{aligned} \tag{1}$$

In the Mid West region, from the 8th to the 18th of August 2024, scheduled care was deferred to focus on unscheduled care. Weeks are referenced by their Monday start date, so the deferral falls in the week beginning Monday the 5th of August 2024. The reset period also has no obvious parametric form, so it is also modelled as a weekly block effect. It is captured with the $\mathcal{MW}_t$ component, which spans the weeks of the Mid West reset period, and three weeks after to capture a return to the regional level. The region gate $\mathbf{1}(j = \text{MW})$ in the mean restricts the effect to the Mid West, so $\mathcal{MW}_t$ is indexed by week alone:

$$\begin{aligned} \mathcal{MW}_t = & \kappa_t \mathbf{1}(t = \text{Aug 5 2024 week}) \\ & + \kappa_t \mathbf{1}(t = \text{Aug 12 2024 week}) \\ & + \kappa_t \mathbf{1}(t = \text{Aug 19 2024 week}) \\ & + \kappa_t \mathbf{1}(t = \text{Aug 26 2024 week}) \\ & + \kappa_t \mathbf{1}(t = \text{Sep 2 2024 week}). \end{aligned} \tag{2}$$

Week-to-week residual autocorrelation is modelled with two autoregressive terms: $\mathcal{E}_{j,t}^1 = \phi_1 (y_{j,t-1} - \mu_{j,t-1})$ at lag 1 and $\mathcal{E}_{j,t}^2 = \phi_2 (y_{j,t-2} - \mu_{j,t-2})$ at lag 2, with ϕ_1 and ϕ_2 shared across regions.

The base model is therefore:

$$\begin{aligned} y_{j,t} &\sim \mathcal{N}(\mu_{j,t} + \mathcal{E}_{j,t}^1 + \mathcal{E}_{j,t}^2, \tau_j^{-1}), \\ \mu_{j,t} &= \alpha_j + \mathcal{C}_{j,t}^{52} + \mathcal{R}_{j,t} + \mathbf{1}(j = \text{MW}) \mathcal{MW}_t. \end{aligned} \tag{3}$$

Secondary cycles are fitted as extensions: 1-month, 2-month, and 6-month cycles ($\mathcal{C}_{j,t}^4$, $\mathcal{C}_{j,t}^8$, and $\mathcal{C}_{j,t}^{26}$ respectively) in Section 5.6.

The addition of components is evaluated with the DIC: each step in Appendix 5 lowers DIC, from 7015 (regional level) to 5336.9 (AR(2)). AR(2) is retained over AR(1) (5336.9 vs 5339.5).

4.4 Bayesian Inference

The previous Section specified the mean structure and its components. What remains is to estimate the parameters with Bayesian inference. Recall: Bayesian inference treats every parameter as a random variable with its own distribution. Section 4.4.1 outlines the likelihood for the data. Section 4.4.2 introduces the prior, the distribution that represents uncertainty before seeing the data. Bayes theorem combines the likelihood, $p(y | \theta)$, and the prior, $p(\theta)$, into a posterior distribution, up to proportionality,

$$p(\theta | y) \propto p(y | \theta) p(\theta),$$

which represents updated uncertainty. The posterior distributions for the parameters represent the uncertainty given the observed data model. This uncertainty is measured by default with 95% credible intervals, and the posterior mean is used as the point estimator.

4.4.1 Likelihood

The first part of the Bayes formula is the likelihood, which represents the probabilistic model for the data. Since the data is aggregated, summing days to weeks and hospitals into regions, the normality assumption becomes more reasonable due to the enlarged counts. The likelihood is therefore modelled as a normal distribution:

$$y_{j,t} \sim \mathcal{N}(\mu_{j,t} + \mathcal{E}_{j,t}^1 + \mathcal{E}_{j,t}^2, \tau_j^{-1}), \quad t > 2,$$

with the lag-1 term only at $t = 2$ and no autoregressive term at $t = 1$. Here $\mu_{j,t}$ and the autoregressive terms are the mean structure developed in Section 4.3. The regional precision is τ_j .

4.4.2 Priors

Priors play a role in the posterior distribution by specifying mean and uncertainty for the parameters before seeing the data. For instance, a subject matter expert might have a strong belief that the annual cycle has a specific amplitude and phase, and could then encode their expectation in a prior distribution. In the case of our report, diffuse priors are used because there is no strong prior belief about any of our parameters; it is preferred to have prior distributions that are guaranteed to play a minimal role in the posterior distributions. The diffuse priors for the parameters are as follows:

$$\begin{aligned} \alpha_j, \beta_j, \gamma_j, \delta_{j,t}, \kappa_t &\sim \mathcal{N}(0, 0.001) \\ \tau_j &\sim \Gamma(0.001, 0.001) \quad \phi_1, \phi_2 \sim \mathcal{N}(0, 0.01). \end{aligned}$$

The normal priors are written in the precision parameterisation used by JAGS, where the second argument is the precision. A precision of 0.001 corresponds to a variance of 1000, approximately equivalent to a standard deviation of ≈ 31.6 . Because our rates range from roughly 0 to 10, the above priors are diffuse. The AR coefficients priors are also diffuse, with a precision of 0.01 (standard deviation of 10). The $\Gamma(0.001, 0.001)$ prior on the precision is diffuse, with a mean of 1 and a standard deviation of ≈ 31.6 . The priors are the same across all regions and weeks, so they do not influence the ranking between regions.

4.4.3 Posterior

Posterior sampling is performed using *Just Another Gibbs Sampler (JAGS)* (Plummer, 2003), implemented by pyJAGS (Nowotny, 2024), a Python interface to JAGS. Sampling uses 4 chains, each sampled independently. Each chain runs 30,000 iterations: the first 10,000 are discarded as burn-in, leaving 20,000 retained per chain. No thinning is applied. Pooling the 4 chains gives 80,000 posterior draws. Convergence is assessed using the Gelman-Rubin diagnostic and trace plots; all R -hat values were at or below 1.0002, with full diagnostics in Appendix 10.9.

4.5 Bayesian Ranking

Health region levels, α_j^s , are used for ranking, where j is one of the 6 HSE health regions, $j \in \{1, \dots, 6\}$, and s is the draw index, $s \in \{1, \dots, n\}$, where the total number of pooled posterior draws is $n = 80,000$ (4 chains $\times$ 20,000 retained).

$$A = \begin{bmatrix} \alpha_1^1 & \cdots & \alpha_j^1 & \cdots & \alpha_6^1 \\ \vdots & & \vdots & & \vdots \\ \alpha_1^s & \cdots & \alpha_j^s & \cdots & \alpha_6^s \\ \vdots & & \vdots & & \vdots \\ \alpha_1^n & \cdots & \alpha_j^n & \cdots & \alpha_6^n \end{bmatrix}.$$

Ranks follow the posterior ranking approach of Laird & Louis (1989). Each posterior sample is ranked to form the rank distribution, R :

$$R = \begin{bmatrix} r_1^1 & \cdots & r_j^1 & \cdots & r_6^1 \\ \vdots & & \vdots & & \vdots \\ r_1^s & \cdots & r_j^s & \cdots & r_6^s \\ \vdots & & \vdots & & \vdots \\ r_1^n & \cdots & r_j^n & \cdots & r_6^n \end{bmatrix},$$

where $r_j^s = \text{rank}(\alpha_j^s) = \sum_{i=1}^6 I(\alpha_j^s > \alpha_i^s)$. When a tie occurs, the average rank is assigned to the tied values.

The posterior mean rank, $\hat{r}_j$, is calculated by taking the mean across the n posterior samples for each region j :

$$\hat{r}_j = \frac{1}{n} \sum_{s=1}^n r_j^s.$$

Collecting the mean ranks across the six regions gives the vector:

$$\hat{R} = (\hat{r}_1, \hat{r}_2, \hat{r}_3, \hat{r}_4, \hat{r}_5, \hat{r}_6).$$

From which the posterior mean rank is ranked, resulting in the ranked mean.

$$R^* = (r_1^*, r_2^*, r_3^*, r_4^*, r_5^*, r_6^*),$$

where $r_j^* = \text{rank}(\hat{r}_j) = \sum_{i=1}^6 I(\hat{r}_j > \hat{r}_i)$.

4.6 Pairwise tests

The ranking orders the regions, but a pairwise test is needed to say whether any pair of regions actually differ on any parameter. Two regions, j and j' , are compared on a parameter θ by subtracting their posteriors, where $j, j' \in \{1, \dots, 6\}$ and $j \neq j'$. Within each draw the difference is $d_{j,j'}^s = \theta_j^s - \theta_{j'}^s$, and the n values give the posterior of the difference, $d_{j,j'}$. Its 95% credible interval is the 2.5th and 97.5th percentiles of $d_{j,j'}^s$. An interval that excludes 0 means the two regions differ at the 95% level. No multiple-comparison correction is applied. The interval describes the difference itself, not a long-run error rate. This is applied to the regional level α_j (Section 5.1.1) and the annual cycle amplitude A_j (Section 5.3.2).

5 Results

In this Section we explore the posterior samples of the parameters, quantities that are derived from those, and model fit diagnostics. The posterior samples are used to calculate the posterior mean and 95% credible intervals (CIs) for the parameters. The posterior mean is the average of the parameter values across all posterior samples, and represents a point estimate of the parameter. The 95% CI is calculated as the 2.5th and 97.5th percentiles of the posterior distribution, and represents the range that has a 95% probability of containing the true parameter value, given the observed data, prior, and model.

The model’s mean component, composed of the level, annual cycle, New Year effect, and Mid West reset effect, is presented first. Followed by model extensions such as the 6, 2, and 1 month cycles. Posterior means, and 95% credible intervals (CIs) are shown below when applicable.

5.1 Mean Component

The mean component of the model, $\mu_{j,t}$, additively captures the regional level, annual cycle, New Year effect, and Mid West reset effect for each region and week. The level differences between regions are captured by the α_j parameters. They are apparent in the aggregate mean component, seen in Figure 8, where the regions with higher levels have higher mean component values across the series.

The annual cycle term peaks in winter and troughs in summer for every region. The New Year effect, a block term, appears as the sharp dips spanning Christmas and the New Year. The Mid West reset, also a block term, appears as the isolated dip in the HSE Mid West component in August 2024. Posterior means and 95% credible intervals for each component are reported below.

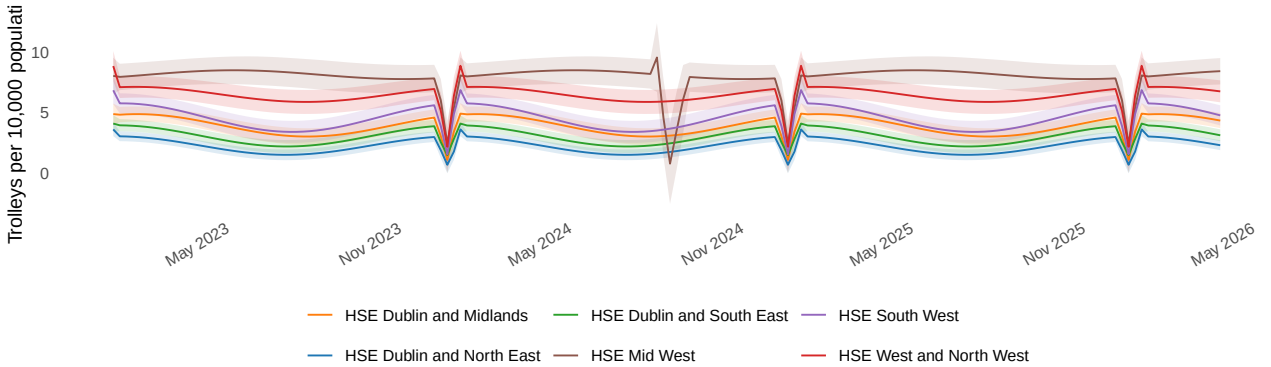


Figure 8: Posterior mean component by week. Colours indicate HSE regions; shaded bands are 95% credible intervals for the mean, $\mu_{j,t}$.

5.1.1 Regional Levels

The regional levels, α_j , capture the expected level for each region; they estimate the rate of the region ignoring any time varying effects. Level parameters are of interest for ranking, as they capture the underlying level for each region, controlling for the other components in the model.

Across regions, the level posterior mean rate per 10,000 population is highest in HSE Mid West (8.13, 95% CI [7.43, 8.83]) and lowest in HSE Dublin and North East (2.27, 95% CI [2.01, 2.53]). All but one pairwise difference between regional levels excludes zero (Appendix Table 7); the exception is HSE South West and HSE Dublin and Midlands, whose levels are not credibly separated. The ordering is otherwise confident. In particular, three western regions (HSE Mid West, HSE South West, HSE West and North West) all have higher levels than the three eastern regions (HSE Dublin and Midlands, HSE Dublin and North East, HSE Dublin and South East), consistent with an East-West divide in demand and service provision. Figure 9 shows the spatial distribution of regional levels, visually confirming East-West differences. Because the levels separate cleanly, their ordering is stable. In the next section this is formalised with the rank distribution.

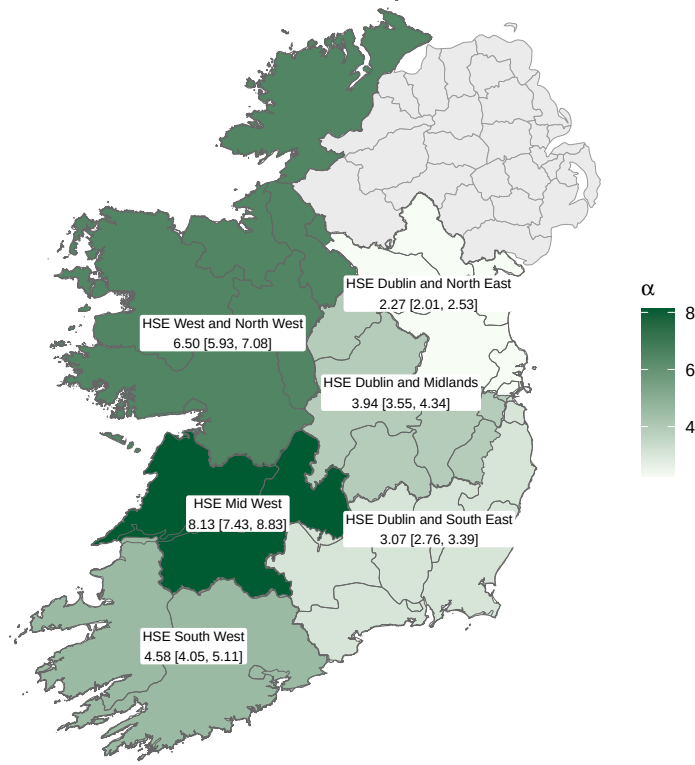


Figure 9: Regional Levels (α_j) per 10,000 population.

5.2 Regional Rankings

The rank distribution follows from Section 4.5 which describes the ranking process. The primary scaling is per 10,000 population; a sensitivity analysis refits the model under three alternative denominators: per 100 inpatient beds, per €1 billion of regional budget, and per 1,000 population aged 65+ to more fully explore the factors impacting the on trolley rates.

5.2.1 Per 10,000 Population

For four of the six regions the mean rank, median rank, ranked mean, and 95% CI coincide on a single integer (Table 1). HSE South West and HSE Dublin and Midlands are the exception, each with a 95% rank CI of [3, 4], swapping ranks 3 and 4 in about 3% of posterior draws. The ordering is HSE Mid West (1), HSE West and North West (2), HSE South West and HSE Dublin and Midlands (ranks 3–4), HSE Dublin and South East (5), HSE Dublin and North East (6). Per 10,000 population still yields the clearest ranking among the four denominators considered, with only the adjacent South West and Dublin and Midlands positions unresolved.

5.2.2 Sensitivity to Scaling

The ranking is broadly robust to scale choice (Table 1). All four scalings agree on HSE Mid West first, HSE West and North West second, and the two Dublin regions at the bottom (HSE Dublin and South East 5th, HSE Dublin and North East 6th). The middle two positions are less stable. Under per €1 billion of regional budget scaling, HSE Dublin and Midlands and HSE Dublin and South East have rank CIs of [4, 5]. Under per 1,000 population aged 65+ scaling, HSE Dublin and Midlands and HSE South West swap into ranks 3 and 4 respectively, both with CIs [3, 4].

Table 1: Posterior ranks (1 = highest alpha) across response scalings. Consistent ranks across sections indicate the ordering is robust to denominator choice. P(off rank) is the posterior probability that a draw places the region at a rank other than its Ranked Mean.

Region	Ranked Mean	Mean Rank	Median Rank	95% CI	P(off rank)
Per 10,000 population					
HSE Dublin and Midlands	4	4.0	4.0	[3, 4]	0.030
HSE Dublin and North East	6	6.0	6.0	[6, 6]	<0.001
HSE Dublin and South East	5	5.0	5.0	[5, 5]	<0.001
HSE Mid West	1	1.0	1.0	[1, 1]	<0.001
HSE South West	3	3.0	3.0	[3, 4]	0.030
HSE West and North West	2	2.0	2.0	[2, 2]	<0.001
Per 100 inpatient beds					
HSE Dublin and Midlands	4	4.0	4.0	[4, 4]	0.002
HSE Dublin and North East	6	6.0	6.0	[6, 6]	<0.001
HSE Dublin and South East	5	5.0	5.0	[5, 5]	0.001
HSE Mid West	1	1.0	1.0	[1, 1]	<0.001
HSE South West	3	3.0	3.0	[3, 3]	0.005
HSE West and North West	2	2.0	2.0	[2, 2]	0.004
Per €1 billion of regional budget					
HSE Dublin and Midlands	4	4.3	4.0	[4, 5]	0.392
HSE Dublin and North East	6	6.0	6.0	[6, 6]	0.012
HSE Dublin and South East	5	4.6	5.0	[4, 5]	0.388
HSE Mid West	1	1.1	1.0	[1, 2]	0.058
HSE South West	3	3.0	3.0	[2, 4]	0.076
HSE West and North West	2	2.0	2.0	[1, 3]	0.089
Per 1,000 population aged 65+					
HSE Dublin and Midlands	3	3.2	3.0	[3, 4]	0.209
HSE Dublin and North East	6	6.0	6.0	[6, 6]	<0.001
HSE Dublin and South East	5	5.0	5.0	[5, 5]	0.001
HSE Mid West	1	1.0	1.0	[1, 1]	<0.001
HSE South West	4	3.8	4.0	[3, 4]	0.209
HSE West and North West	2	2.0	2.0	[2, 2]	<0.001

5.3 Annual Cycle

In this section, the annual cycle is analysed in terms of its phase and amplitude. The phase captures the timing of the cycle, while the amplitude captures the strength of the cycle. Both are derived from the cosine and sine coefficients of the annual cycle, β_j and γ_j , as described in Section 4.3. This section continues to use the primary scaling of per 10,000 population. Sensitivity analyses for the annual cycle are not performed, as the cycle is a relatively small effect that should be less sensitive to scaling than the level and ranking.

5.3.1 Annual Cycle Phase

Four of six regions have 95% CI widths narrower than 3 months: HSE Dublin and Midlands (10.0 weeks), HSE Dublin and North East (7.6 weeks), HSE Dublin and South East (8.1 weeks), and HSE South West (10.4 weeks). HSE West and North West (25.9 weeks) and HSE Mid West (44.1 weeks) are wider. The four narrow CIs cluster close to the New Year, seen in Figure 10, indicating a consistent annual cycle with winter peaks.

Five of the six regions place most of their posterior mass on a peak in winter, where winter spans December to February (Met Éireann, 2026). The winter peak is near-certain for HSE Dublin and North East (99.7%), HSE Dublin and South East (99.5%), and HSE South West (97.9%), and clear but less concentrated for HSE Dublin and Midlands (93.3%) and HSE West and North West (71.6%). HSE Mid West (12.9%) is the

Table 2: annual cycle amplitude.

Region	Mean	95% CI	Contains 0
HSE Dublin and Midlands	0.97	[0.47, 1.49]	No
HSE Dublin and North East	0.79	[0.45, 1.13]	No
HSE Dublin and South East	0.90	[0.50, 1.32]	No
HSE Mid West	0.67	[0.12, 1.45]	No
HSE South West	1.24	[0.57, 1.92]	No
HSE West and North West	0.76	[0.16, 1.46]	No

exception, with its peak poorly identified, a wide credible interval, and less pronounced amplitude (Table 2). Credible intervals, probabilities, and posterior means can be found in Appendix Table 4.

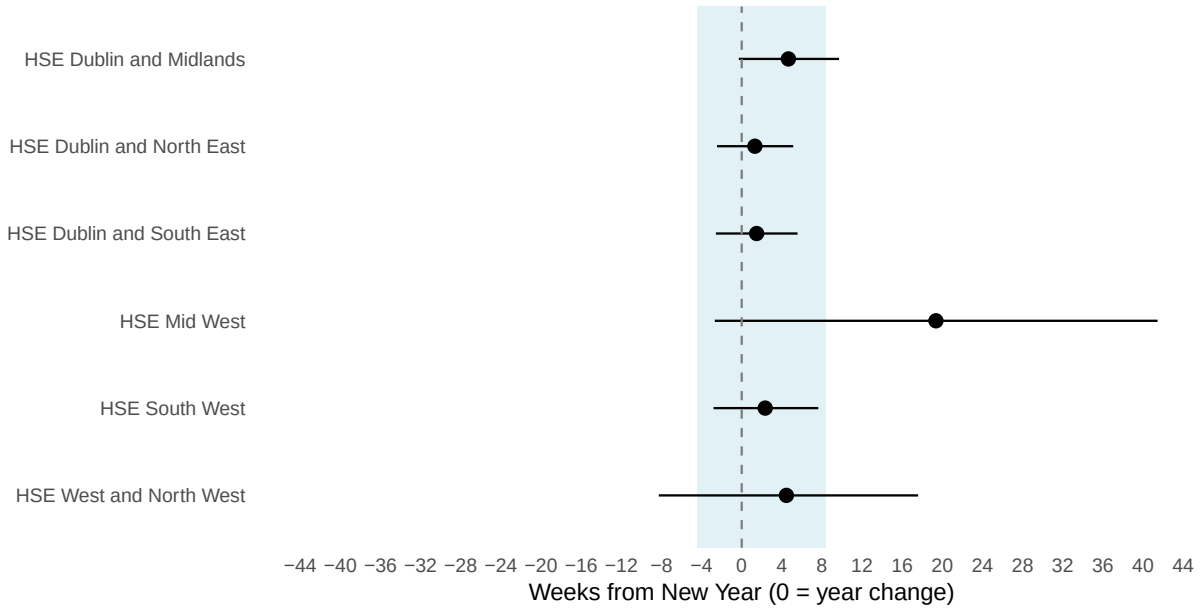


Figure 10: Peak phases plot. Shaded blue region represents winter (1st of December to end of February). Black points are the regional posterior means. Bars are the 95% credible interval.

5.3.2 Annual Cycle Amplitude

The posterior means of the amplitude range from 0.62 in HSE Mid West to 1.24 in HSE South West. The amplitude posterior means add a large trolley burden to some regions. For instance, the cycle in HSE Dublin and North East swings $\pm 35\%$ (95% CI $[\pm 20\%, \pm 51\%]$) around the regional level: winter peaks sit that fraction above the regional mean and summer troughs sit the same fraction below. All 95% credible intervals are wide. The pairwise test in Appendix Table 8 shows the regions are not comparable as all region combinations have a 95% CI which contains 0.

The model comparison agrees. A single shared cycle has a slightly lower DIC than the region-specific cycle (Appendix Table 6). The regions differ little in their annual cycle. The amplitudes are not separated and five of six phases cluster near the winter peak. HSE Mid West is the exception, with the widest phase interval and the least pronounced amplitude, so its annual cycle cannot be resolved.

5.4 New Year Effect

Based on Figure 11, the New Year effect on all regions has a roughly similar shape. HSE Dublin and North East and HSE Dublin and South East have the narrowest CIs, suggesting their New Year effects are the most consistent across observed years. During the New Year week, regions differ in whether their credible intervals return to the regional level. Some include it, others exclude it. For instance HSE South West (-0.88 [-2.19, 0.44]), and HSE West and North West (-0.69 [-2.12, 0.76]) return to the regional level. One week after the New Year, HSE Dublin and North East (0.57 [0.07, 1.07]), HSE South West (1.10 [0.08, 2.12]), and HSE West and North West (1.77 [0.66, 2.88]) have CI lower bounds above the level. The remaining three span the level. HSE West and North West has the highest posterior mean.

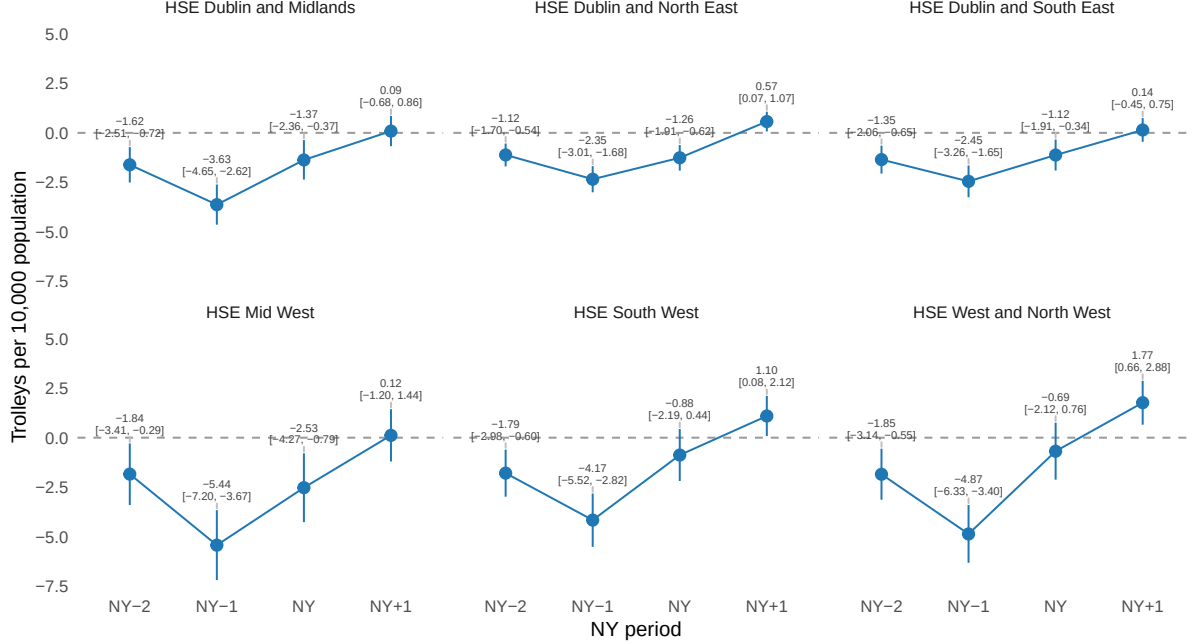


Figure 11: New Year effect by region. Points are posterior means; bars are 95% credible intervals. Each point is annotated with its posterior mean and 95% CI. Dashed line at zero, representing the regional level

5.5 Mid West Reset

The HSE Mid West deferral period runs from the 8th to the 18th of August 2024, falling within the weeks of the 5th, 1.40 [-1.28, 4.11], and the 12th of August, -3.57 [-6.71, -0.43], shown in Figure 12. By posterior mean, the minimum is the week of the 19th of August, -7.29 [-10.57, -4.02]. The week of the 2nd of September, -1.88 [-4.59, 0.82], shows a return to the regional level.

5.6 Model Extensions

The base model in Equation (3) is extended with three secondary cycles: a 4 week cycle, an 8 week cycle, and a 26 week cycle. As discussed in the spectral analysis (Section 3.4.3), the periodogram in Figure 5 shows elevated power near the 4, 8, and 26 week periods across regions; the extensions are specified in Section 4.3. By Table 3, the 4 and 8 week cycles decrease deviance but increase DIC; the components fit better but add too much complexity. The 26 week cycle increases both deviance and DIC; meaning the additional component worsens fit. None of the cycles lower DIC below the base model. AR(2) was selected as the base over AR(1) on the same criterion (DIC 5336.9 vs 5339.5); the full development sequence is in Appendix 5.

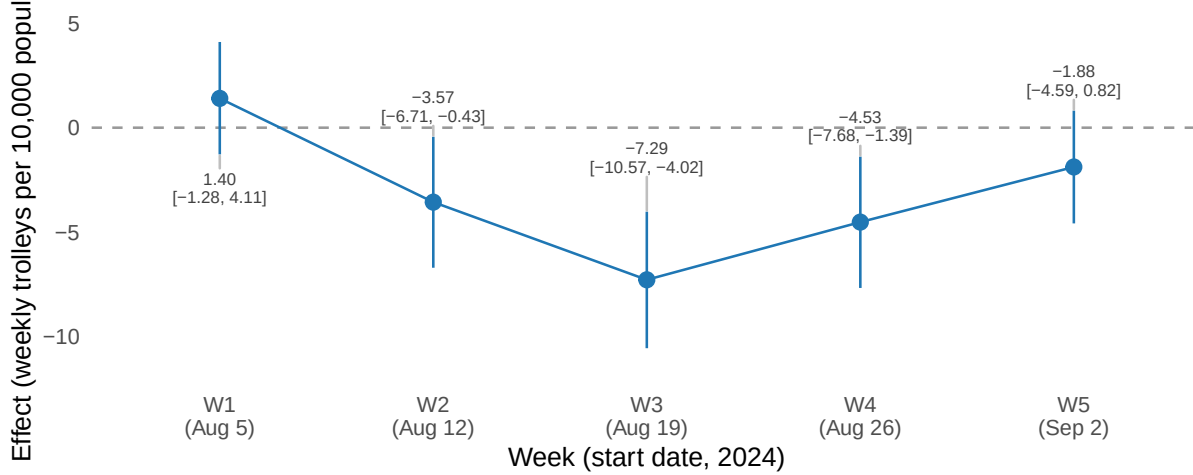


Figure 12: Mid West reset effect for the five weeks from the 5th of August 2024. Points are posterior means, bars are 95% credible intervals, with each point annotated with its mean and 95% CI. Dashed line at zero.

Table 3: Cycle extensions added to the base model (per 10,000 population).

Description	Deviance	pD	DIC
Base model	5225.0	111.8	5336.9
Base + 4 week cycle	5220.4	136.3	5356.7
Base + 8 week cycle	5212.9	136.3	5349.2
Base + 26 week cycle	5240.3	136.2	5376.6

5.7 Residuals

The residual is the gap between the observation and the full one-step prediction, $r_{j,t} = y_{j,t} - (\mu_{j,t} + \mathcal{E}_{j,t}^1 + \mathcal{E}_{j,t}^2)$. Since the observation variance is τ_j^{-1} , dividing by the standard deviation, $\tau_j^{-1/2}$, gives the standardised residual

$$z_{j,t} = r_{j,t} \sqrt{\tau_j}.$$

Under a well-specified model the residuals already have mean 0. Dividing by the standard deviation $\tau_j^{-1/2}$ is what brings them to variance 1, on a common scale across regions.

Shown in Figure 13, the model residuals are not white noise. There are clear cycles, however, the 4, 8 and 26 week cycles tested in Section 5.6 do not improve model fit as measured by the DIC. There is visible regional dependence which can be seen when regions move in step; appearing to synchronise. This is shown to be meaningful in the following section, 5.7.1.

5.7.1 Cross-Correlation

Figure 14 (below) shows a heatmap of 0-lags of the cross-correlation function (CCF) results of residual region pairs. The CCF works by extending the ACF, $\rho_{xy}(k) = \text{corr}(x_t, y_{t-k})$, where x is one region's residual signal and y is another. All 15 region pairs have a cross-correlation at lag-0 exceeding the significance band, which provides evidence for a regional dependence structure.

We expect regional dependence structure to be driven by demographics like age, with Dublin and non-Dublin regions clustering separately. The data only partly supports this. The three strongest pairs are Dublin and Midlands with West and North West (0.34), Dublin and Midlands with Dublin and North East (0.32), and Dublin and North East with West and North West (0.30). Two of the three cross the Dublin and non-Dublin boundary, so the dependence structure does not split cleanly along Dublin lines. We would also

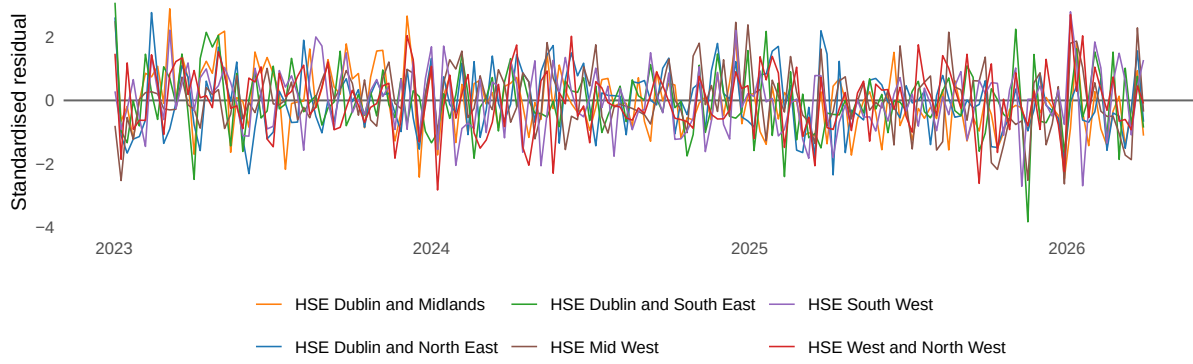


Figure 13: Standardised posterior mean residuals from the selected model, by week. Colour indicates region.

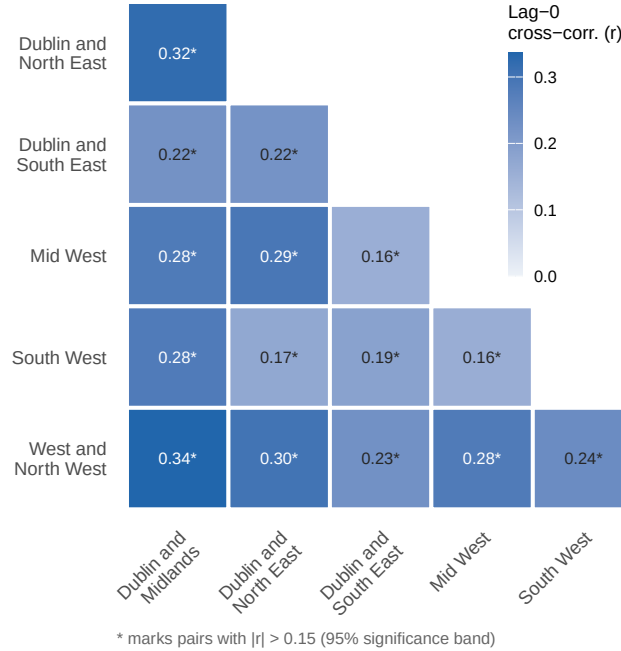


Figure 14: Lag-0 cross-correlation of posterior mean residuals for all 15 HSE region pairs. Darker blue indicates a stronger positive correlation. An asterisk marks pairs above the 95% significance band ($|r| > 0.15$).

expect weaker correlations between regions that do not share a border, however, HSE South West and West and North West have a moderate correlation (0.24) despite sharing no border. The reverse also holds. South West correlates least with its actual neighbours, Mid West (0.16) and Dublin and South East (0.19), the two weakest links in the matrix. Geographic adjacency does not explain the dependence structure.

6 Discussion

This study applies a Bayesian time series regression framework to weekly hospital trolley rates across the six HSE health regions of Ireland, with the primary aim of comparing regional burden after controlling for population size, seasonality, and known operational events. The regression component, capturing regional levels, an annual cycle, New Year effects, and a Mid West operational reset, is the central focus of this analysis. The autoregressive term serves a secondary role, accounting for residual week-to-week persistence in the data rather than being of substantive interest in itself. An AR(2) term is selected: it gives the lowest DIC (5336.9 against 5339.5 for the AR(1) specification), and the lag-2 coefficient ϕ_2 is credibly non-zero (0.08, 95% CI [0.02, 0.15]). Three further extensions, a 4-week cycle, an 8-week cycle, and a 26-week cycle,

were tested and not adopted, as all worsened DIC. The selected model therefore comprises regional levels, an annual cycle, region-specific New Year effects, a Mid West operational reset, and an AR(2) persistence term.

The most substantive finding is the magnitude and credibility of regional differences in trolley pressure levels. The level posterior mean rate per 10,000 population is highest in HSE Mid West (8.13 [7.43, 8.83]) and lowest in HSE Dublin and North East (2.27 [2.01, 2.53]). The pairwise comparison of posterior level distributions reveals near-certain regional orderings: for almost all region pairs, the posterior probability that one region has a higher level than another is either effectively 0 or 1, indicating the estimated ordering is not an artefact of posterior uncertainty. Fourteen of the fifteen pairwise credible intervals exclude zero (Appendix Table 7); the single exception is HSE South West and HSE Dublin and Midlands, whose levels are not credibly separated. A clear west/east divide is evident, with the three western regions consistently showing substantially higher rates than the three Dublin regions. This divide is invisible in raw regional totals (Figure 15) because larger regions, particularly the Dublin regions, have substantially larger populations, producing higher absolute counts that mask the underlying per-capita pressure. Population standardisation is therefore essential for any meaningful comparison across regions of different sizes within a nationally administered health system.

The west/east divide likely reflects structural features of the Irish health system and population geography. HSE Mid West, the highest-burden region, covers a large and predominantly rural catchment spanning counties Limerick, Clare, and Tipperary, with acute hospital services concentrated in a small number of centres. The spatial distribution of the population across this region, combined with relatively limited acute capacity relative to catchment size, creates conditions where patients are more likely to experience delays in admission. A similar dynamic applies, to varying degrees, across the other western regions. Dublin regions, by contrast, benefit from a higher concentration of acute bed capacity relative to their populations, which may facilitate more rapid patient flow through emergency departments.

The formal Bayesian ranking of regions by their level, derived from the posterior samples of the regional levels in the spirit of Laird & Louis (1989), yields a stable and near-deterministic ordering. HSE Dublin and North East ranks lowest in burden and HSE Mid West highest, with negligible posterior standard deviation on the rank estimates. Under the primary per 10,000 population scaling, rank credible intervals collapse to a single integer for four of the six regions. The exception is the adjacent pair HSE South West and HSE Dublin and Midlands, which swap ranks 3 and 4 in about 3% of draws, each with a 95% CI of [3, 4]. This ordering is robust across all four scaling denominators examined: per 10,000 population, per 100 inpatient beds, per €1 billion of regional budget, and per 1,000 population aged 65+. The same South West and Dublin and Midlands pair again carries a 95% CI of [3, 4] under the age-structure scaling, confining the instability to these adjacent middle positions. The stability of the top two and bottom two positions across all denominators reinforces confidence in the principal finding.

An annual cycle is identified across all six regions, with five of six placing substantial posterior mass on a winter peak within two months of New Year (Figure 10, Appendix Table 4). This is consistent with established patterns of respiratory illness demand in Ireland and internationally, where cold weather and increased viral transmission drive hospital admissions in the winter months. HSE Mid West is the notable outlier, with only 12.9% of posterior mass in the winter window and a wide 95% CI on peak phase spanning 44.1 weeks. This less clearly defined, low amplitude cycle in the Mid West may reflect the influence of the operational reset period in August 2024 on the estimated cycle for that region. Regional differences in cycle amplitude, while suggestive, cannot be credibly distinguished given the available data, as all pairwise amplitude comparisons contain zero (Appendix Table 8).

The New Year dip is identified across all regions and is consistent with the known HSE practice of discharging patients during the Christmas period in anticipation of the January surge in respiratory illness demand. The region-specific New Year effect, which allows each region to respond differently across the four-week window, captures genuine heterogeneity in the timing and magnitude of this effect that a global specification could not accommodate. The subsequent rebound above the regional level in several regions in the week after New Year reflects the rapid return of demand once the holiday period ends. The HSE Mid West operational reset in August 2024 produces a clear and localised dip in trolley rates, reaching a minimum in the week of the 19th of August (-7.29 [-10.57, -4.02] per 10,000), with return to the regional level

by early September. The effect is clinically plausible and statistically well identified, with credible intervals excluding zero across the reset weeks. The absence of any comparable pattern in the other five regions over the same period confirms this is a region-specific effect attributable to the operational intervention rather than a seasonal or system-wide phenomenon.

The standardised residuals are approximately normal across regions, with only mild heavy tails in HSE Dublin and South East, supporting the Normal likelihood (Appendix Figures 16, 17). Residual diagnostics indicate that some sub-annual structure and cross-regional dependence remain unexplained by the selected model (Appendix Figures 18 and 19). Significant cross-correlations at lag zero are observed across all 15 region pairs, suggesting that some shared temporal variation is not captured by the current independent-regions specification.

7 Limitations and Future Work

The model operates on weekly aggregated trolley rates, which smooths over the pronounced daily variation visible in the partial autocorrelation function. A model operating at daily resolution with an explicit within-week cycle would provide a more granular characterisation of hospital pressure dynamics and may be of greater operational relevance to healthcare managers. Weekly binning also splits the New Year holiday unevenly, as the 1st of January falls on a different weekday each year (Section 3.4.1), so daily counts would give more observations to identify the New Year and Mid West reset effects. The weekly aggregation also means that short-term spikes in pressure, for example following a single large influx of respiratory admissions, are partially obscured in the regional totals. Post-weekend and post-public holiday surges in trolley counts are a documented feature of the Irish hospital system, driven by reduced discharge rates over weekends and bank holidays (Department of Health, 2025). While the New Year effect is explicitly modelled, other public holidays such as Easter and bank holidays generate comparable patterns of pre-holiday discharge and post-holiday surge that are not separately accounted for in the current framework.

The model treats each HSE region as independent, with no mechanism for shared temporal variation across regions. The residual cross-correlations observed across all 15 region pairs suggest this assumption is violated in practice. Extensions incorporating a multivariate structure, shared latent component, or spatial random effect could better account for system-wide shocks that affect all regions simultaneously, such as national respiratory illness surges or policy changes, and would be a natural direction for future work.

The AR(2) persistence parameter ϕ is shared across all six regions, which may not fully capture heterogeneity in the speed at which congestion dissipates across different regional health systems. It is plausible that regions with greater bed capacity or more developed discharge infrastructure recover more quickly from elevated pressure, a question that a region-specific AR(2) extension could address. Such an extension would however require careful consideration of identifiability given the length of the available series.

The Mid West reset term was motivated by a known system-wide overcrowding response implemented in the region in August 2024, and its inclusion was therefore justified on operational grounds prior to model fitting. However, the model does not systematically test whether comparable structural breaks occurred in other regions during the observation period, nor does it account for other unobserved system-level interventions that may have affected trolley rates. Extending the framework to detect and model such events across all regions, rather than specifying them a priori, represents a useful direction for future work.

The observation period of approximately three years, while sufficient to identify the main seasonal and calendar features of the data, limits the ability to assess longer-run trends in hospital pressure or to evaluate the stability of the regional ranking over multiple years. As the HSE health region structure was only established in 2023, extending the analysis to future data will be important for confirming whether the west/east divide identified here reflects a stable structural feature of the system or is partly an artefact of the transition period.

A further limitation is that the analysis does not account for structural covariates that may explain some of the observed regional variation, such as differences in bed capacity, staffing levels, population age structure, external facilities that can alleviate pressure or socioeconomic deprivation. Incorporating such covariates

within a Bayesian regression framework would allow the analysis to distinguish between variation in trolley pressure attributable to measurable system characteristics and variation that reflects more fundamental differences in regional demand relative to capacity.

The estimated peaks cluster in the weeks around the New Year (Appendix Table 4). A shared phase with region-specific amplitude could lead to a more parsimonious model, testing whether regions differ in the strength of the winter peak rather than its timing.

The annual cycle is currently a calendar term with no external driver. Winter trolley peaks coincide with respiratory illness admissions. Supplementing the cycle with a respiratory surveillance covariate, such as weekly influenza or RSV positivity from national surveillance, would move the model from describing the winter peak to explaining it. It would also let the peak respond to severity that varies year to year.

The model is fit to describe and rank the regions, not to predict. The same structure of regional level, annual cycle, and autoregressive terms can produce short-term forecasts of trolley counts. Forecasts would need out-of-sample evaluation with proper scoring rules, rather than the in-sample DIC used here. A validated forecast would give the HSE advance warning of regional surges.

The trolley counts come from a single source, the HSE daily report. The Irish Nurses and Midwives Organisation publishes an independently collected daily trolley count. Comparing the two series would test the reliability of the reported figures and flag hospitals where the counts diverge, such as the under-reporting observed at one Model 4 hospital.

8 Conclusion

This study develops a Bayesian time series framework for analysing weekly hospital trolley rates across the six HSE health regions of Ireland. Starting from a regional level with an autoregressive component, the model is extended iteratively to include an annual cycle, region-specific New Year block effects, and a Mid West operational reset term for August to September 2024. The selected specification uses an AR(2) persistence term; three sub-annual cycle extensions were tested but did not improve DIC. The model captures the dominant temporal features of the data within a parsimonious framework.

The central finding is a pronounced and statistically credible west/east divide in hospital pressure levels, with HSE Mid West carrying the highest burden and HSE Dublin and North East the lowest. This ordering is robust across all four scaling denominators examined (Table 1) and is supported by near-certain pairwise posterior comparisons across fourteen of the fifteen region pairs. These findings are entirely invisible in raw regional totals and emerge only through population standardisation, reinforcing the importance of per capita scaling when making regional comparisons within a nationally administered health system. Five of six regions exhibit a winter-peaking annual cycle consistent with respiratory illness demand, and both the New Year dip and the Mid West operational reset are clearly identified as distinct features of the data.

By embedding these dynamics within a Bayesian framework, the analysis propagates uncertainty through to all model parameters and provides a principled basis for probabilistic regional comparison. Future work could explore daily-resolution modelling, region-specific AR parameters, and the incorporation of structural covariates such as bed capacity, staffing, and population age structure to help explain the observed regional disparities. As the HSE six-region structure was only established in 2023, extending the analysis as additional data accumulate will be important for confirming whether the west/east divide is a stable structural feature of the Irish hospital system. More broadly, this study demonstrates that Bayesian time series modelling of routinely collected administrative data can yield substantive, policy relevant insights into the distribution of hospital pressure across a national health system, supporting more evidence-informed approaches to capacity planning and resource allocation. An interactive version of these results is available at hsereport.ie.

9 References

- Asplin, B. R., Magid, D. J., Rhodes, K. V., Solberg, L. I., Lurie, N., & Camargo, C. A. (2003). A conceptual model of emergency department crowding. *Annals of Emergency Medicine*, 42(2), 173–180. <https://doi.org/10.1067/mem.2003.302>
- Austin, P. C. (2002). A comparison of Bayesian methods for profiling hospital performance. *Medical Decision Making*, 22(2), 163–172. <https://doi.org/10.1177/0272989X0202200213>
- Bowers, F. (2026). *Risk to patient care at UHL 'intolerable' - Medical Board*. <https://www.rte.ie/news/munter/2026/0511/1572742-university-hospital-limerick/>
- Brick, A., & Kakoulidou, T. (2025). *Projections of regional demand and bed capacity requirements for public acute hospitals in Ireland, 2023–2040: Based on the Hippocrates model* [Report]. ESRI. <https://doi.org/10.26504/sustat132>
- Central Statistics Office. (2023). *FY006B: Population by single year of age, census of population 2022 (PxStat table)*. <https://data.cso.ie/table/FY006B>
- Central Statistics Office. (2024a). *HHR01: HSE health region populations, census 2022 (PxStat table)*. <https://data.cso.ie/table/HHR01>
- Central Statistics Office. (2024b). *HHR02: Demographic profile of HSE health regions (PxStat table)*. <https://data.cso.ie/table/HHR02>
- Central Statistics Office. (2026). *DHA74: Total hospital discharges by age group (PxStat table)*. <https://data.cso.ie/table/DHA74>
- Department of Health. (2023). *Minister for Health thanks healthcare workers for delivering 20% reduction in trolley numbers in last 6 months of 2023*. <https://www.gov.ie/en/department-of-health/press-releases/minister-for-health-thanks-healthcare-workers-for-delivering-20-reduction-in-trolley-numbers-in-last-6-months-of-2023/>
- Department of Health. (2025). *Minister Carroll MacNeill focuses on further reducing Emergency Department overcrowding and enhancing services nationwide*. <https://www.gov.ie/en/department-of-health/press-releases/minister-carroll-macneill-focuses-on-further-reducing-emergency-department-overcrowding-and-enhancing-services-nationwide/>
- Gelman, A. (2004). Exploratory data analysis for complex models. *Journal of Computational and Graphical Statistics*, 13(4). <https://doi.org/10.1198/106186004X11435>
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman & Hall/CRC. <https://doi.org/10.1201/b16018>
- Health Information and Quality Authority. (2025). *Advice to the Minister for Health to inform decision-making around the design and delivery of urgent and emergency healthcare services in the HSE Mid West region of Ireland*. Health Information; Quality Authority. <https://www.hiqa.ie/reports-and-publications/key-reports-investigations/advice-minister-health-inform-decision-making>
- Health Service Executive. (2025). *Reply to Parliamentary Question 46162/25 (Deputy Martin Daly): Total in-patient and day-case beds in each public and voluntary hospital*. Parliamentary Question reply, Dáil Éireann.
- Health Service Executive. (2026a). *Urgent and emergency care report*. <https://www2.hse.ie/services/urgent-emergency-care-report/>
- Health Service Executive. (2026b). *HSE reports 448 people currently hospitalised with respiratory illness*. <https://about.hse.ie/news/hse-reports-448-people-currently-hospitalised-with-respiratory-illness/>
- Irish Nurses and Midwives Organisation. (2024). *122,186 patients treated on trolleys in 2024*. <https://www.inmo.ie/News-Campaigns/Details/122186-patients-treated-on-trolleys-in-2024>
- Jacob, N., Chalkley, M., Santos, R., & Siciliani, L. (2024). Variation in attendance at emergency departments in England across local areas: A system under unequal pressure. *Health Policy*, 150, 105186. <https://doi.org/10.1016/j.healthpol.2024.105186>
- Laird, N. M., & Louis, T. A. (1989). Empirical Bayes ranking methods. *Journal of Educational Statistics*, 14(1), 29–46. <https://doi.org/10.3102/10769986014001029>
- Lynch, B., Browne, J., Buckley, C. M., Healy, O., Corcoran, P., & Fitzgerald, A. P. (2019). An interrupted time-series analysis of the impact of emergency department reconfiguration on regional emergency department trolley numbers in Ireland from 2005 to 2015. *BMJ Open*, 9(9), e029261. <https://doi.org/10.1136/bmjopen-2019-029261>

- MacDermott, C., Scarrott, C. J., & Ferguson, J. (2025). *Bayes-ically fair: A Bayesian Ranking of the Olympic Medal Table*. <https://doi.org/10.48550/arXiv.2510.14723>
- Mellor, J. et al. (2023). Forecasting influenza hospital admissions within English sub-regions using hierarchical generalised additive models. *Communications Medicine*, 3, 190. <https://doi.org/10.1038/s43856-023-00424-4>
- Met Éireann. (2026). *Climate of Ireland*. The Irish Meteorological Service. <https://www.met.ie/climate/climate-of-ireland>
- Morley, C., Unwin, M., Peterson, G. M., Stankovich, J., & Kinsman, L. (2018). Emergency department crowding: A systematic review of causes, consequences and solutions. *PLOS ONE*, 13(8), e0203316. <https://doi.org/10.1371/journal.pone.0203316>
- Mulroe, J., Donohue, F., Kavanagh, P. M., McCarthy, S., & Johnson, H. (2019). A comparison of summer and winter emergency hospitalisations in Ireland. *Irish Medical Journal*, 112(5), 935. <https://imj.ie/a-comparison-of-summer-and-winter-emergency-hospitalisations-in-ireland/>
- Nowotny, M. (2024). *pyJAGS: A Python interface to JAGS* (Version 2.4.0) [Computer software]. <https://github.com/michaelnowotny/pyjags>
- PE Global. (2026). *Navigating NCHD Changeovers: Tips and Support*. Locum Express. <https://www.locumexpress.ie/january-and-july-changeover-advice/>
- Plummer, M. (2003). JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*. <https://www.r-project.org/conferences/DSC-2003/Proceedings/Plummer.pdf>
- RTÉ. (2023). *Why do the HSE and INMO trolley figures differ?* RTÉ News, Analysis and Comment. <https://www.rte.ie/news/analysis-and-comment/2023/0112/1346239-trolley-count/>
- Sahu, S. K., Baffour, B., Harper, P. R., Minty, J. H., & Sarra, C. (2014). A hierarchical Bayesian model for improving short-term forecasting of hospital demand by including meteorological information. *Journal of the Royal Statistical Society Series A*, 177(1), 39–61. <https://doi.org/10.1111/rssa.12008>
- Shumway, R. H., & Stoffer, D. S. (2017). *Time series analysis and its applications: With R examples* (4th ed.). Springer. <https://doi.org/10.1007/978-3-319-52452-8>
- Smith, S. W. (1999). *The scientist and engineer's guide to digital signal processing* (2nd ed.). California Technical Publishing. <https://dspguide.com/>
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., & Linde, A. van der. (2002). Bayesian measures of model complexity and fit. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(4). <https://doi.org/10.1111/1467-9868.00353>
- Upshur, R. E. G., Moineddin, R., Crighton, E., Kiefer, L., & Mamdani, M. (2005). Simplicity within complexity: Seasonality and predictability of hospital admissions in the province of Ontario 1988–2001, a population-based analysis. *BMC Health Services Research*, 5, 13. <https://doi.org/10.1186/1472-6963-5-13>

10 Appendix

10.1 Annual Cycle Peak Phase

Table 4: Peak week of the annual cycle relative to New Year

Region	Mean peak (wk from NY)	95% CI (wk from NY)	P(winter)
HSE Dublin and Midlands	4.7	[-0.3, +9.7]	0.933
HSE Dublin and North East	1.3	[-2.5, +5.1]	0.997
HSE Dublin and South East	1.5	[-2.6, +5.6]	0.995
HSE Mid West	19.4	[-2.7, +41.4]	0.129
HSE South West	2.3	[-2.8, +7.6]	0.979
HSE West and North West	4.5	[-8.3, +17.6]	0.716

10.2 Model Development DIC

The model is built up one component at a time. In general, each is retained when it lowers DIC.

Table 5: Model development sequence: DIC as each component is added (per 10,000 population).

Description	Deviance	pD	DIC
Regional level	6990.7	24.3	7015.0
+ Annual cycle	6702.3	48.5	6750.8
+ Regional New Year	6271.2	97.0	6368.3
+ Mid West reset	6215.1	107.2	6322.2
+ AR(1)	5229.6	109.9	5339.5
+ AR(2)	5225.0	111.8	5336.9

10.3 Results of Regional Structure Changes

Each ablation is the reported model with one feature collapsed to a single value shared across all regions. This isolates the value of the regional level and the regional annual cycle inside the reported model.

Table 6: Regional-structure ablations of the reported model. Each row is the reported model with one feature collapsed to a single value shared across regions. DIC change is the ablation minus the reported model. A positive change means the regional feature lowers DIC and is justified. A negative change means a shared feature fits as well or better.

Model	Deviance	pD	DIC	DIC change
Reported model (regional level, regional cycle)	5225.0	111.8	5336.9	
Global level (level shared across regions)	5436.1	102.1	5538.2	+201.3
Shared annual cycle (cycle shared across regions)	5220.6	91.7	5312.3	-24.6

10.4 Pairwise Regional Level Comparisons

Each row is the posterior difference between two regional levels (α_j), per 10,000 population. An interval that excludes 0 means the two regions differ at the 95% level.

Table 7: Pairwise comparison of regional levels (α_j) per 10,000 population.

Comparison	vs	Mean diff	95% CI	Contains 0
HSE Mid West	HSE Dublin and North East	5.86	[5.11, 6.60]	No
HSE Mid West	HSE Dublin and South East	5.06	[4.28, 5.83]	No
HSE West and North West	HSE Dublin and North East	4.23	[3.60, 4.87]	No
HSE Mid West	HSE Dublin and Midlands	4.19	[3.38, 4.99]	No
HSE Mid West	HSE South West	3.55	[2.67, 4.43]	No
HSE West and North West	HSE Dublin and South East	3.43	[2.77, 4.08]	No
HSE West and North West	HSE Dublin and Midlands	2.56	[1.86, 3.26]	No
HSE South West	HSE Dublin and North East	2.31	[1.72, 2.89]	No
HSE West and North West	HSE South West	1.92	[1.14, 2.70]	No
HSE Dublin and Midlands	HSE Dublin and North East	1.67	[1.20, 2.15]	No
HSE Mid West	HSE West and North West	1.63	[0.72, 2.54]	No
HSE South West	HSE Dublin and South East	1.51	[0.90, 2.12]	No
HSE Dublin and Midlands	HSE Dublin and South East	0.87	[0.37, 1.38]	No
HSE Dublin and South East	HSE Dublin and North East	0.80	[0.40, 1.21]	No
HSE South West	HSE Dublin and Midlands	0.64	[-0.03, 1.29]	Yes

10.5 Pairwise Cycle Amplitude Comparisons

Each row is the posterior difference between two regional annual cycle amplitudes (A_j), per 10,000 population. Every interval contains 0, so no two regions are credibly separated on amplitude.

Table 8: Annual cycle amplitude pairwise

Comparison	vs	Mean diff	95% CI	Contains 0
HSE South West	HSE Mid West	0.57	[-0.44, 1.47]	Yes
HSE South West	HSE West and North West	0.48	[-0.49, 1.39]	Yes
HSE South West	HSE Dublin and North East	0.45	[-0.30, 1.21]	Yes
HSE South West	HSE Dublin and South East	0.33	[-0.46, 1.13]	Yes
HSE Dublin and Midlands	HSE Mid West	0.30	[-0.61, 1.08]	Yes
HSE South West	HSE Dublin and Midlands	0.27	[-0.57, 1.11]	Yes
HSE Dublin and South East	HSE Mid West	0.23	[-0.63, 0.94]	Yes
HSE Dublin and Midlands	HSE West and North West	0.21	[-0.65, 1.00]	Yes
HSE Dublin and Midlands	HSE Dublin and North East	0.18	[-0.42, 0.79]	Yes
HSE Dublin and South East	HSE West and North West	0.15	[-0.66, 0.87]	Yes
HSE Dublin and North East	HSE Mid West	0.12	[-0.72, 0.78]	Yes
HSE Dublin and South East	HSE Dublin and North East	0.12	[-0.41, 0.64]	Yes
HSE West and North West	HSE Mid West	0.09	[-0.88, 1.00]	Yes
HSE Dublin and Midlands	HSE Dublin and South East	0.07	[-0.58, 0.72]	Yes
HSE Dublin and North East	HSE West and North West	0.03	[-0.74, 0.72]	Yes

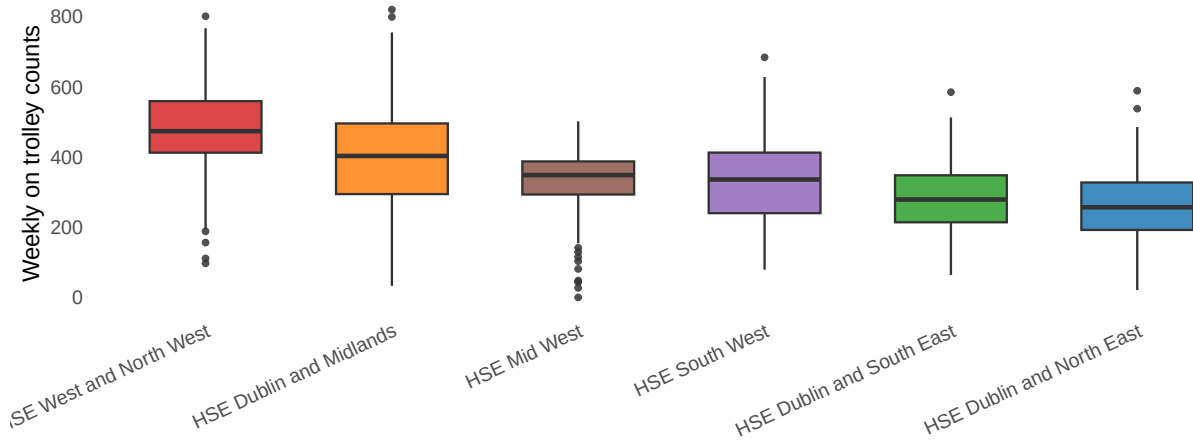


Figure 15: Distribution of weekly on trolley counts by HSE region across the full series (Jan 2023 – Apr 2026). Box shows the IQR, central line the median; points beyond are individual weeks.

10.6 Raw Weekly Counts by Region

10.7 Residual Normality Checks

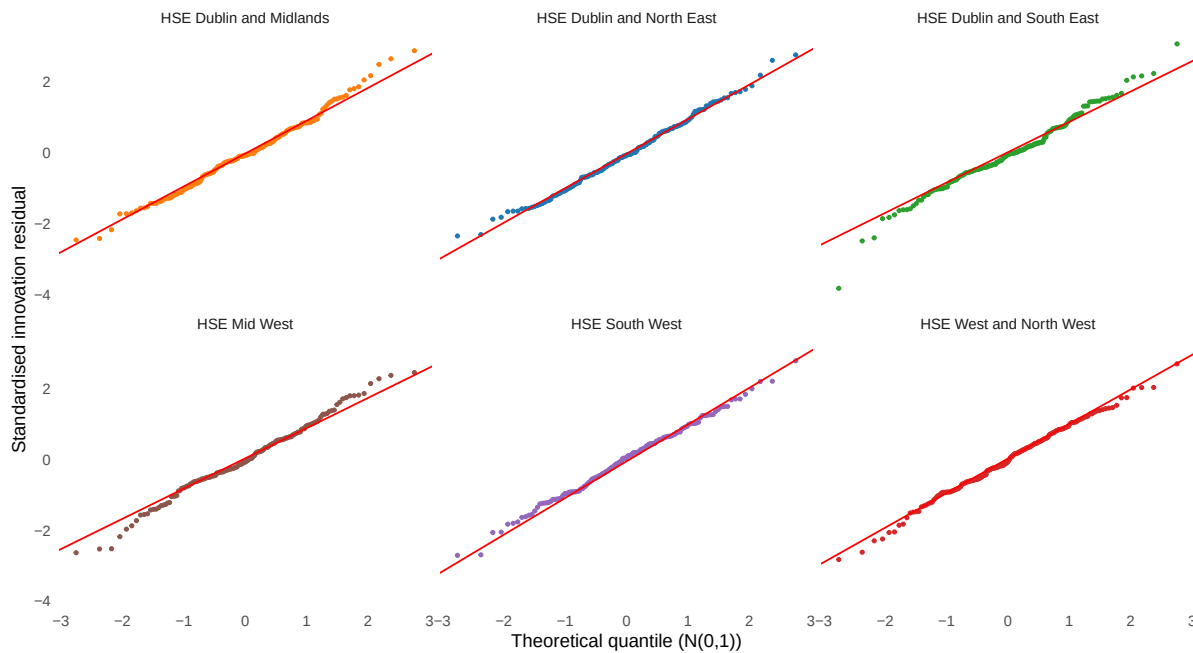


Figure 16: Normal quantile-quantile plots of the standardised residuals by region. Points on the red line indicate normality.

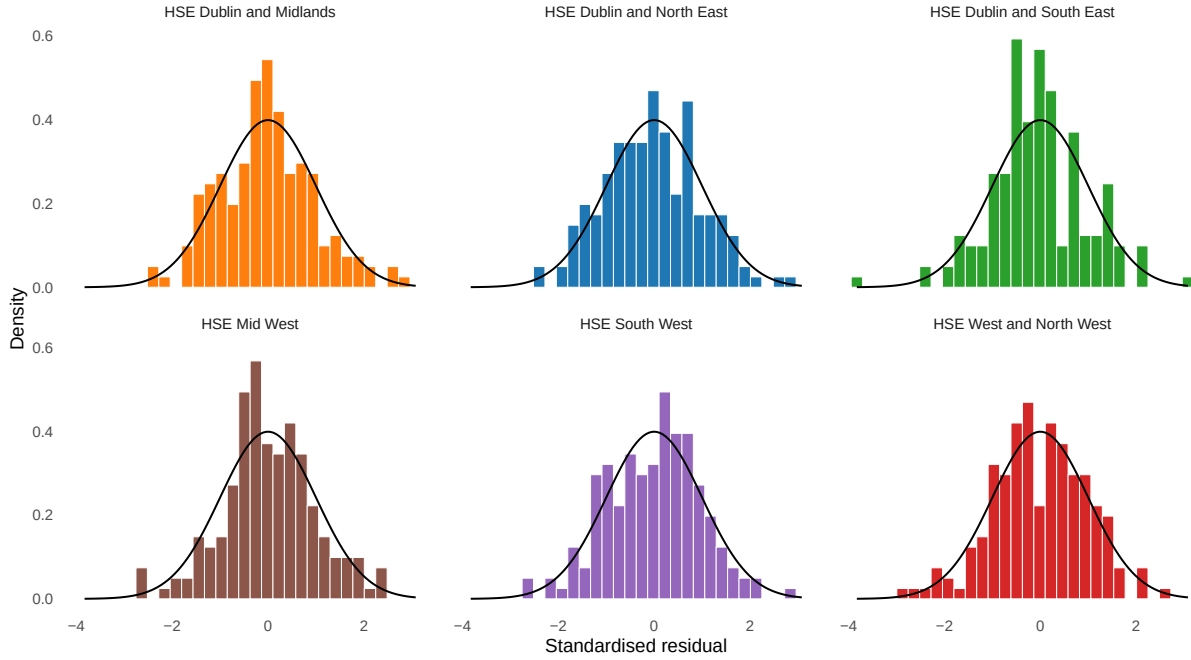


Figure 17: Histograms of the standardised residuals by region with the standard normal density overlaid.

10.8 Residual Cycles

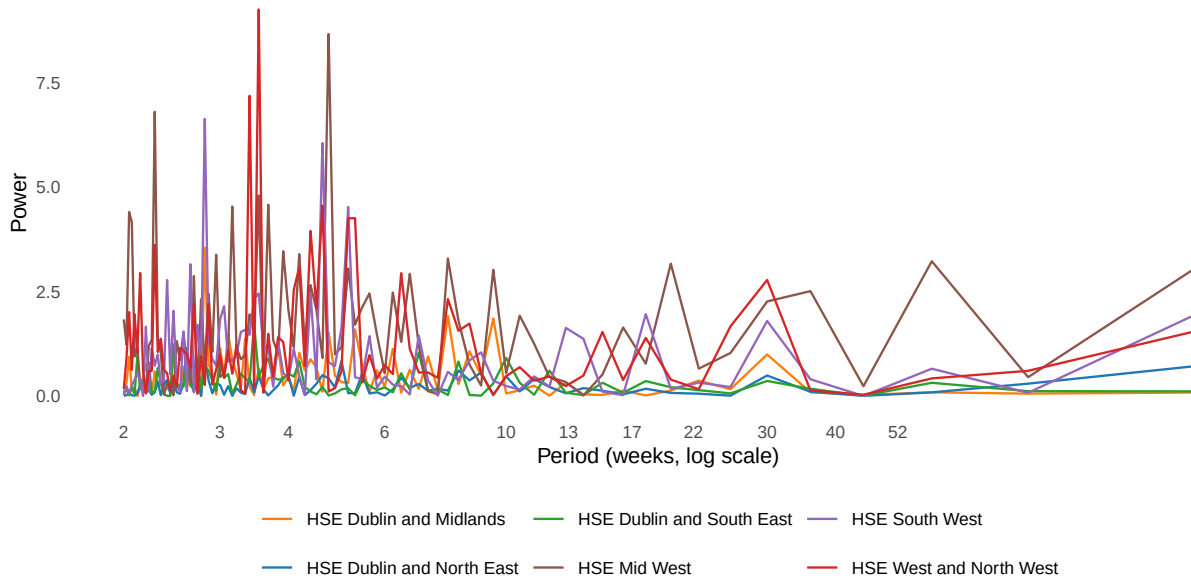


Figure 18: Power spectral density of the posterior-mean innovation residuals by region (period in weeks, log scale). Peaks mark periodic structure remaining after the annual cycle, New Year, and Mid West reset terms and the AR(2) correction.

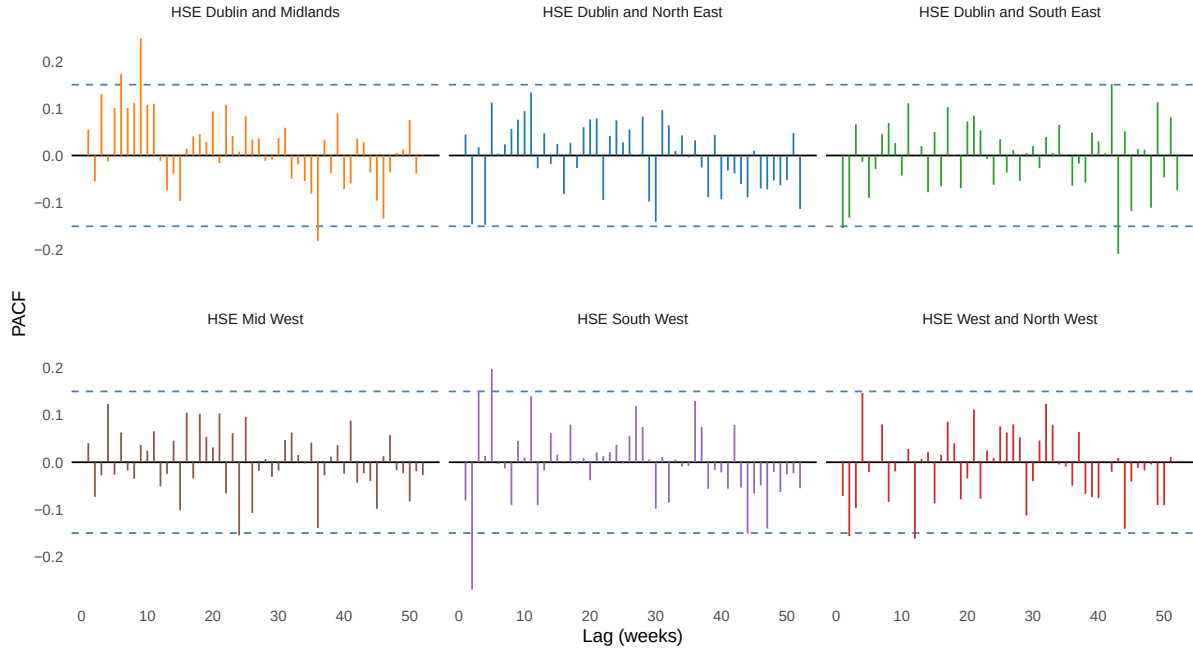


Figure 19: Partial autocorrelation of the posterior-mean innovation residuals by region, to lag 52 weeks. Dashed bands are the 95% white-noise bounds. Bars beyond them mark week-to-week dependence not captured by the AR(2) term.

10.9 Trace Plots and Convergence

Convergence diagnostics for the selected model (AR(2) with the date-anchored New Year effect). Trace plots show all monitored parameters across the four chains after burn-in. The Gelman-Rubin table reports the potential scale reduction factor (R-hat) for every parameter.

Throughout this section the bracketed index on a parameter denotes the HSE region: 1 Dublin and Midlands, 2 Dublin and North East, 3 Dublin and South East, 4 Mid West, 5 South West, 6 West and North West. The four New Year window effects are numbered in temporal order: `delta_1` the week two weeks before the New Year week, `delta_2` the week before, `delta_3` the New Year week (containing Jan 1), and `delta_4` the week after.

Table 9: Gelman-Rubin potential scale reduction factor (R-hat) for every monitored parameter of the selected model. Values at or near 1.00 indicate convergence; the maximum observed was 1.0002.

Parameter	R-hat	Parameter	R-hat
alpha[1]	1.0000	delta_3[5]	1.0002
alpha[2]	1.0000	delta_3[6]	1.0001
alpha[3]	1.0000	delta_4[1]	1.0002
alpha[4]	1.0000	delta_4[2]	1.0001
alpha[5]	1.0000	delta_4[3]	1.0001
alpha[6]	1.0000	delta_4[4]	1.0001
beta[1]	1.0000	delta_4[5]	1.0000
beta[2]	1.0001	delta_4[6]	1.0000
beta[3]	1.0001	gamma[1]	1.0001
beta[4]	1.0000	gamma[2]	1.0001
beta[5]	1.0000	gamma[3]	1.0001
beta[6]	1.0000	gamma[4]	1.0000
delta_1[1]	1.0000	gamma[5]	1.0000
delta_1[2]	1.0001	gamma[6]	1.0000
delta_1[3]	1.0000	phi1	1.0001
delta_1[4]	1.0001	phi2	1.0001
delta_1[5]	1.0001	sigma_w1	1.0001
delta_1[6]	1.0001	sigma_w2	1.0001
delta_2[1]	1.0001	sigma_w3	1.0002
delta_2[2]	1.0001	sigma_w4	1.0002
delta_2[3]	1.0001	sigma_w5	1.0001
delta_2[4]	1.0001	tau[1]	1.0000
delta_2[5]	1.0003	tau[2]	1.0000
delta_2[6]	1.0001	tau[3]	1.0000
delta_3[1]	1.0001	tau[4]	1.0000
delta_3[2]	1.0001	tau[5]	1.0000
delta_3[3]	1.0001	tau[6]	1.0000
delta_3[4]	1.0001		